

## gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs Library

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| Cell Groups                        |
|------------------------------------|
| GF180MCU_OSU_SC_GP9T3V3__ADDF_1    |
| GF180MCU_OSU_SC_GP9T3V3__ADDH_1    |
| GF180MCU_OSU_SC_GP9T3V3__AND2_1    |
| GF180MCU_OSU_SC_GP9T3V3__AOI21_1   |
| GF180MCU_OSU_SC_GP9T3V3__AOI22_1   |
| GF180MCU_OSU_SC_GP9T3V3__BUF_16    |
| GF180MCU_OSU_SC_GP9T3V3__BUF_1     |
| GF180MCU_OSU_SC_GP9T3V3__BUF_2     |
| GF180MCU_OSU_SC_GP9T3V3__BUF_4     |
| GF180MCU_OSU_SC_GP9T3V3__BUF_8     |
| GF180MCU_OSU_SC_GP9T3V3__CLKBUF_16 |
| GF180MCU_OSU_SC_GP9T3V3__CLKBUF_1  |
| GF180MCU_OSU_SC_GP9T3V3__CLKBUF_2  |
| GF180MCU_OSU_SC_GP9T3V3__CLKBUF_4  |
| GF180MCU_OSU_SC_GP9T3V3__CLKBUF_8  |
| GF180MCU_OSU_SC_GP9T3V3__CLKINV_16 |
| GF180MCU_OSU_SC_GP9T3V3__CLKINV_1  |
| GF180MCU_OSU_SC_GP9T3V3__CLKINV_2  |
| GF180MCU_OSU_SC_GP9T3V3__CLKINV_4  |
| GF180MCU_OSU_SC_GP9T3V3__CLKINV_8  |
| GF180MCU_OSU_SC_GP9T3V3__DECAP_1   |
| GF180MCU_OSU_SC_GP9T3V3__DFFN_1    |
| GF180MCU_OSU_SC_GP9T3V3__DFFSR_1   |

|                                  |
|----------------------------------|
| GF180MCU_OSU_SC_GP9T3V3__DFF_1   |
| GF180MCU_OSU_SC_GP9T3V3__DLATN_1 |
| GF180MCU_OSU_SC_GP9T3V3__DLAT_1  |
| GF180MCU_OSU_SC_GP9T3V3__INV_16  |
| GF180MCU_OSU_SC_GP9T3V3__INV_1   |
| GF180MCU_OSU_SC_GP9T3V3__INV_2   |
| GF180MCU_OSU_SC_GP9T3V3__INV_4   |
| GF180MCU_OSU_SC_GP9T3V3__INV_8   |
| GF180MCU_OSU_SC_GP9T3V3__MUX2_1  |
| GF180MCU_OSU_SC_GP9T3V3__NAND2_1 |
| GF180MCU_OSU_SC_GP9T3V3__NOR2_1  |
| GF180MCU_OSU_SC_GP9T3V3__OAI21_1 |
| GF180MCU_OSU_SC_GP9T3V3__OAI22_1 |
| GF180MCU_OSU_SC_GP9T3V3__OAI31_1 |
| GF180MCU_OSU_SC_GP9T3V3__OR2_1   |
| GF180MCU_OSU_SC_GP9T3V3__TBUF_1  |
| GF180MCU_OSU_SC_GP9T3V3__TIEH    |
| GF180MCU_OSU_SC_GP9T3V3__TIEL    |
| GF180MCU_OSU_SC_GP9T3V3__TINV_1  |
| GF180MCU_OSU_SC_GP9T3V3__XNOR2_1 |
| GF180MCU_OSU_SC_GP9T3V3__XOR2_1  |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_ADDF\_1

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT |   |    | OUTPUT |     |
|-------|---|----|--------|-----|
| A     | B | CI | CO     | SUM |
| 0     | 0 | 0  | 0      | 0   |
| 0     | 0 | 1  | 0      | 1   |
| 0     | 1 | 0  | 0      | 1   |
| 0     | 1 | 1  | 1      | 0   |
| 1     | 0 | 0  | 0      | 1   |
| 1     | 0 | 1  | 1      | 0   |
| 1     | 1 | 0  | 1      | 0   |
| 1     | 1 | 1  | 1      | 1   |

## Footprint

| Cell Name                       | Area     |
|---------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__addf_1 | 88.83000 |

## Pin Capacitance Information

| Cell Name                       | Pin Cap(pf) |         |         | Max Cap(pf) |          |
|---------------------------------|-------------|---------|---------|-------------|----------|
|                                 | A           | B       | CI      | CO          | SUM      |
| gf180mcu_osu_sc_gp9t3v3__addf_1 | 0.01546     | 0.01477 | 0.01140 | 15.45427    | 15.49830 |

## Leakage Information

| Cell Name                       | Leakage(nW) |         |         |
|---------------------------------|-------------|---------|---------|
|                                 | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__addf_1 | 0.00000     | 0.00428 | 0.00452 |

## Delay Information

Delay(ns) to CO rising :

| Cell Name                       | Timing Arc(Dir) | Delay(ns) |         |          |
|---------------------------------|-----------------|-----------|---------|----------|
|                                 |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__addf_1 | A->CO (RR)      | 0.16388   | 1.94506 | 67.96260 |
|                                 | B->CO (RR)      | 0.18243   | 2.23710 | 72.00240 |
|                                 | CI->CO (RR)     | 0.16417   | 1.98160 | 66.05660 |

Delay(ns) to CO falling :

| Cell Name                       | Timing Arc(Dir) | Delay(ns) |         |          |
|---------------------------------|-----------------|-----------|---------|----------|
|                                 |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__addf_1 | A->CO (FF)      | 0.19281   | 2.59811 | 75.87250 |
|                                 | B->CO (FF)      | 0.18762   | 2.92298 | 80.44350 |
|                                 | CI->CO (FF)     | 0.16286   | 2.76821 | 76.43470 |

Delay(ns) to SUM rising :

| Cell Name                       | Timing Arc(Dir) | Delay(ns) |         |          |
|---------------------------------|-----------------|-----------|---------|----------|
|                                 |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__addf_1 | A->SUM (-R)     | 0.33779   | 2.69218 | 77.22690 |
|                                 | B->SUM (-R)     | 0.33078   | 3.11285 | 83.92300 |
|                                 | CI->SUM (-R)    | 0.27743   | 2.87212 | 79.14260 |

Delay(ns) to SUM falling :

| Cell Name                       | Timing Arc(Dir) | Delay(ns) |         |          |
|---------------------------------|-----------------|-----------|---------|----------|
|                                 |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__addf_1 | A->SUM (-F)     | 0.19853   | 3.15590 | 85.27720 |
|                                 | B->SUM (-F)     | 0.23217   | 2.98178 | 82.54350 |
|                                 | CI->SUM (-F)    | 0.24200   | 2.73937 | 79.28210 |

## Power Information

Internal switching power(pJ) to CO rising :

| Cell Name                       | Input | Power(pJ) |         |         |
|---------------------------------|-------|-----------|---------|---------|
|                                 |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__addf_1 | A     | 0.02475   | 0.19994 | 3.32863 |
|                                 | A     | 0.06694   | 0.23643 | 3.36524 |
|                                 | B     | 0.03179   | 0.18050 | 3.00361 |
|                                 | B     | 0.07278   | 0.22173 | 3.04686 |
|                                 | CI    | 0.02543   | 0.16429 | 2.60811 |
|                                 | CI    | 0.05411   | 0.19299 | 2.63823 |

Internal switching power(pJ) to CO falling :

| Cell Name                       | Input | Power(pJ) |         |         |
|---------------------------------|-------|-----------|---------|---------|
|                                 |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__addf_1 | A     | 0.07837   | 0.24779 | 3.36938 |
|                                 | A     | 0.03611   | 0.20858 | 3.32971 |
|                                 | B     | 0.07180   | 0.22537 | 3.02910 |
|                                 | B     | 0.02982   | 0.18400 | 2.98795 |
|                                 | CI    | 0.06534   | 0.20812 | 2.65833 |
|                                 | CI    | 0.03206   | 0.17501 | 2.62494 |

Internal switching power(pJ) to SUM rising :

| Cell Name                       | Input | Power(pJ) |         |         |
|---------------------------------|-------|-----------|---------|---------|
|                                 |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__addf_1 | A     | 0.00547   | 0.24896 | 4.97816 |
|                                 | A     | 0.08990   | 0.33352 | 5.04754 |
|                                 | B     | 0.01057   | 0.28175 | 5.39195 |
|                                 | B     | 0.09181   | 0.36278 | 5.46719 |
|                                 | CI    | 0.01791   | 0.32509 | 6.10489 |
|                                 | CI    | 0.09491   | 0.40113 | 6.17698 |

**Internal switching power(pJ) to SUM falling :**

| Cell Name                       | Input | Power(pJ) |         |         |
|---------------------------------|-------|-----------|---------|---------|
|                                 |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__addf_1 | A     | 0.09698   | 0.34816 | 5.05153 |
|                                 | A     | 0.00991   | 0.26158 | 4.97638 |
|                                 | B     | 0.09098   | 0.36428 | 5.46441 |
|                                 | B     | 0.01582   | 0.28934 | 5.40967 |
|                                 | CI    | 0.09568   | 0.40590 | 6.23518 |
|                                 | CI    | 0.03045   | 0.34180 | 6.17445 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_ADDH\_1

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT |   | OUTPUT |     |
|-------|---|--------|-----|
| A     | B | CO     | SUM |
| 0     | 0 | 0      | 0   |
| 0     | 1 | 0      | 1   |
| 1     | 0 | 0      | 1   |
| 1     | 1 | 1      | 0   |

## Footprint

| Cell Name                       | Area     |
|---------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__addh_1 | 54.81000 |

## Pin Capacitance Information

| Cell Name                       | Pin Cap(pf) |         | Max Cap(pf) |          |
|---------------------------------|-------------|---------|-------------|----------|
|                                 | A           | B       | CO          | SUM      |
| gf180mcu_osu_sc_gp9t3v3__addh_1 | 0.00768     | 0.00697 | 15.63292    | 15.55909 |

## Leakage Information

| Cell Name                       | Leakage(nW) |         |         |
|---------------------------------|-------------|---------|---------|
|                                 | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__addh_1 | 0.00000     | 0.00342 | 0.00370 |

## Delay Information

Delay(ns) to CO rising :

| Cell Name                       | Timing Arc(Dir) | Delay(ns) |         |          |
|---------------------------------|-----------------|-----------|---------|----------|
|                                 |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__addh_1 | A->CO (RR)      | 0.12992   | 1.98777 | 70.23780 |
|                                 | B->CO (RR)      | 0.12803   | 2.22976 | 73.94310 |

Delay(ns) to CO falling :

| Cell Name                       | Timing Arc(Dir) | Delay(ns) |         |          |
|---------------------------------|-----------------|-----------|---------|----------|
|                                 |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__addh_1 | A->CO (FF)      | 0.11392   | 2.39449 | 73.28180 |
|                                 | B->CO (FF)      | 0.10507   | 2.15559 | 68.80000 |

Delay(ns) to SUM rising (conditional):

| Cell Name                       | Timing Arc(Dir) | When | Delay(ns) |         |          |
|---------------------------------|-----------------|------|-----------|---------|----------|
|                                 |                 |      | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__addh_1 | A->SUM (RR)     | !B   | 0.13152   | 2.11468 | 72.13070 |
|                                 | A->SUM (FR)     | B    | 0.19246   | 2.50599 | 76.09100 |
|                                 | B->SUM (RR)     | !A   | 0.10627   | 1.76512 | 65.99510 |
|                                 | B->SUM (FR)     | A    | 0.20643   | 2.26058 | 71.03440 |

Delay(ns) to SUM falling (conditional):

| Cell Name                       | Timing Arc(Dir) | When | Delay(ns) |         |          |
|---------------------------------|-----------------|------|-----------|---------|----------|
|                                 |                 |      | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__addh_1 | A->SUM (FF)     | !B   | 0.13699   | 2.26205 | 71.51870 |
|                                 | A->SUM (RF)     | B    | 0.20801   | 1.69403 | 56.90440 |
|                                 | B->SUM (FF)     | !A   | 0.12194   | 2.55697 | 76.30660 |
|                                 | B->SUM (RF)     | A    | 0.20584   | 2.00043 | 62.07060 |



## Power Information

Internal switching power(pJ) to CO rising :

| Cell Name                       | Input | Power(pJ) |         |         |
|---------------------------------|-------|-----------|---------|---------|
|                                 |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__addh_1 | A     | 0.03124   | 0.21473 | 3.49101 |
|                                 | A     | 0.04954   | 0.23291 | 3.51037 |
|                                 | B     | 0.03597   | 0.20608 | 3.17358 |
|                                 | B     | 0.04788   | 0.21711 | 3.18540 |

Internal switching power(pJ) to CO falling :

| Cell Name                       | Input | Power(pJ) |         |         |
|---------------------------------|-------|-----------|---------|---------|
|                                 |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__addh_1 | A     | 0.05033   | 0.23943 | 3.53090 |
|                                 | A     | 0.03197   | 0.22112 | 3.51237 |
|                                 | B     | 0.04873   | 0.21694 | 3.17362 |
|                                 | B     | 0.03753   | 0.20584 | 3.16270 |

Internal switching power(pJ) to SUM rising (conditional):

| Cell Name                       | Input | When | Power(pJ) |         |         |
|---------------------------------|-------|------|-----------|---------|---------|
|                                 |       |      | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__addh_1 | A     | B    | 0.05035   | 0.23976 | 3.53721 |
|                                 | A     | B    | 0.03199   | 0.22142 | 3.51944 |
|                                 | A     | !B   | 0.01420   | 0.30523 | 5.62210 |
|                                 | A     | !B   | 0.06632   | 0.35722 | 5.66498 |
|                                 | B     | A    | 0.04876   | 0.21689 | 3.17887 |
|                                 | B     | A    | 0.03756   | 0.20580 | 3.16821 |
|                                 | B     | !A   | 0.00843   | 0.26495 | 4.87956 |
|                                 | B     | !A   | 0.04636   | 0.30253 | 4.91193 |

Internal switching power(pJ) to SUM falling (conditional):

| Cell Name                       | Input | When | Power(pJ) |         |         |
|---------------------------------|-------|------|-----------|---------|---------|
|                                 |       |      | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__addh_1 | A     | B    | 0.03121   | 0.21435 | 3.48511 |
|                                 | A     | B    | 0.04951   | 0.23258 | 3.50453 |
|                                 | A     | !B   | 0.06323   | 0.35340 | 5.67216 |
|                                 | A     | !B   | 0.01119   | 0.30176 | 5.62349 |
|                                 | B     | A    | 0.03596   | 0.20522 | 3.16867 |
|                                 | B     | A    | 0.04787   | 0.21620 | 3.18330 |
|                                 | B     | !A   | 0.05473   | 0.31138 | 4.90910 |
|                                 | B     | !A   | 0.01620   | 0.27365 | 4.87267 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_AND2\_1

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT |   | OUTPUT |
|-------|---|--------|
| A     | B | Y      |
| 0     | x | 0      |
| 1     | 0 | 0      |
| 1     | 1 | 1      |

## Footprint

| Cell Name                       | Area     |
|---------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__and2_1 | 26.46000 |

## Pin Capacitance Information

| Cell Name                       | Pin Cap(pf) |         | Max Cap(pf) |
|---------------------------------|-------------|---------|-------------|
|                                 | A           | B       | Y           |
| gf180mcu_osu_sc_gp9t3v3__and2_1 | 0.00404     | 0.00402 | 15.46230    |

## Leakage Information

| Cell Name                       | Leakage(nW) |         |         |
|---------------------------------|-------------|---------|---------|
|                                 | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__and2_1 | 0.00000     | 0.00144 | 0.00205 |

## Delay Information

Delay(ns) to Y rising :

| Cell Name                       | Timing Arc(Dir) | Delay(ns) |         |          |
|---------------------------------|-----------------|-----------|---------|----------|
|                                 |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__and2_1 | A->Y (RR)       | 0.09992   | 2.10608 | 72.94700 |
|                                 | B->Y (RR)       | 0.10147   | 1.88267 | 69.30150 |

Delay(ns) to Y falling :

| Cell Name                       | Timing Arc(Dir) | Delay(ns) |         |          |
|---------------------------------|-----------------|-----------|---------|----------|
|                                 |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__and2_1 | A->Y (FF)       | 0.08556   | 2.03273 | 67.88340 |
|                                 | B->Y (FF)       | 0.09497   | 2.28660 | 72.38090 |

## Power Information

Internal switching power(pJ) to Y rising :

| Cell Name                       | Input | Power(pJ) |         |         |
|---------------------------------|-------|-----------|---------|---------|
|                                 |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__and2_1 | A     | 0.01761   | 0.33396 | 5.93597 |
|                                 | A     | 0.04077   | 0.35693 | 5.95045 |
|                                 | B     | 0.01677   | 0.36168 | 6.57348 |
|                                 | B     | 0.04503   | 0.38967 | 6.59631 |

Internal switching power(pJ) to Y falling :

| Cell Name                       | Input | Power(pJ) |         |         |
|---------------------------------|-------|-----------|---------|---------|
|                                 |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__and2_1 | A     | 0.03788   | 0.35415 | 5.95264 |
|                                 | A     | 0.01472   | 0.33161 | 5.93904 |
|                                 | B     | 0.04653   | 0.39679 | 6.59413 |
|                                 | B     | 0.01822   | 0.36909 | 6.56596 |

Passive power(pJ) for A rising (conditional):

| Cell Name                       | When      | Power(pJ) |          |          |
|---------------------------------|-----------|-----------|----------|----------|
|                                 |           | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__and2_1 | (!B * !Y) | -0.01399  | -0.01399 | -0.01410 |
|                                 | (!B * !Y) | 0.00180   | 0.00181  | 0.00173  |

Passive power(pJ) for A falling (conditional):

| Cell Name                       | When      | Power(pJ) |          |          |
|---------------------------------|-----------|-----------|----------|----------|
|                                 |           | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__and2_1 | (!B * !Y) | 0.01426   | 0.01426  | 0.01420  |
|                                 | (!B * !Y) | -0.00170  | -0.00168 | -0.00167 |

Passive power(pJ) for B rising (conditional):

| Cell Name                       | When      | Power(pJ) |          |          |
|---------------------------------|-----------|-----------|----------|----------|
|                                 |           | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__and2_1 | (!A * !Y) | -0.01352  | -0.01357 | -0.01348 |
|                                 | (!A * !Y) | 0.00651   | 0.00655  | 0.00642  |

Passive power(pJ) for B falling (conditional):

| Cell Name                       | When      | Power(pJ) |          |          |
|---------------------------------|-----------|-----------|----------|----------|
|                                 |           | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__and2_1 | (!A * !Y) | 0.01380   | 0.01388  | 0.01357  |
|                                 | (!A * !Y) | -0.00624  | -0.00652 | -0.00642 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_AOI21\_1

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT |    |   | OUTPUT |
|-------|----|---|--------|
| A0    | A1 | B | Y      |
| 0     | x  | 0 | 1      |
| x     | x  | 1 | 0      |
| 1     | 0  | 0 | 1      |
| 1     | 1  | x | 0      |

## Footprint

| Cell Name                        | Area     |
|----------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__aoi21_1 | 24.57000 |

## Pin Capacitance Information

| Cell Name                        | Pin Cap(pf) |         |         | Max Cap(pf) |
|----------------------------------|-------------|---------|---------|-------------|
|                                  | A0          | A1      | B       | Y           |
| gf180mcu_osu_sc_gp9t3v3__aoi21_1 | 0.00396     | 0.00398 | 0.00405 | 7.79615     |

## Leakage Information

| Cell Name                        | Leakage(nW) |         |         |
|----------------------------------|-------------|---------|---------|
|                                  | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__aoi21_1 | 0.00000     | 0.00094 | 0.00180 |

## Delay Information

Delay(ns) to Y rising :

| Cell Name                        | Timing Arc(Dir) | Delay(ns) |         |          |
|----------------------------------|-----------------|-----------|---------|----------|
|                                  |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__aoi21_1 | A0->Y (FR)      | 0.09719   | 3.22983 | 85.09710 |
|                                  | A1->Y (FR)      | 0.07870   | 3.16927 | 84.48700 |
|                                  | B->Y (FR)       | 0.07648   | 3.92719 | 98.03850 |

Delay(ns) to Y falling :

| Cell Name                        | Timing Arc(Dir) | Delay(ns) |         |          |
|----------------------------------|-----------------|-----------|---------|----------|
|                                  |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__aoi21_1 | A0->Y (RF)      | 0.07488   | 2.16937 | 60.56870 |
|                                  | A1->Y (RF)      | 0.07264   | 2.73644 | 72.46930 |
|                                  | B->Y (RF)       | 0.03490   | 1.92392 | 53.23420 |



## Power Information

Internal switching power(pJ) to Y rising :

| Cell Name                        | Input | Power(pJ) |         |         |
|----------------------------------|-------|-----------|---------|---------|
|                                  |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__aoi21_1 | A0    | 0.04064   | 0.20888 | 2.52993 |
|                                  | A0    | 0.00267   | 0.17059 | 2.48938 |
|                                  | A1    | 0.03148   | 0.18747 | 2.32616 |
|                                  | A1    | -0.00134  | 0.15383 | 2.29211 |
|                                  | B     | 0.02194   | 0.21499 | 2.79248 |
|                                  | B     | -0.00050  | 0.19189 | 2.76960 |

Internal switching power(pJ) to Y falling :

| Cell Name                        | Input | Power(pJ) |         |         |
|----------------------------------|-------|-----------|---------|---------|
|                                  |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__aoi21_1 | A0    | 0.00428   | 0.16281 | 2.28175 |
|                                  | A0    | 0.04216   | 0.20094 | 2.31956 |
|                                  | A1    | 0.00476   | 0.14962 | 2.00645 |
|                                  | A1    | 0.03753   | 0.18326 | 2.06764 |
|                                  | B     | -0.00383  | 0.17383 | 2.54048 |
|                                  | B     | 0.01868   | 0.19652 | 2.60354 |

Passive power(pJ) for A0 rising (conditional):

| Cell Name                        | When           | Power(pJ) |          |          |
|----------------------------------|----------------|-----------|----------|----------|
|                                  |                | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__aoi21_1 | (A1 * B * !Y)  | -0.01272  | -0.01338 | -0.01322 |
|                                  | (A1 * B * !Y)  | 0.00690   | 0.00655  | 0.00652  |
|                                  | (!A1 * B * !Y) | -0.01352  | -0.01355 | -0.01351 |
|                                  | (!A1 * B * !Y) | 0.00653   | 0.00648  | 0.00645  |
|                                  | (!A1 * !B * Y) | -0.01345  | -0.01351 | -0.01351 |
|                                  | (!A1 * !B * Y) | 0.00643   | 0.00645  | 0.00645  |

Passive power(pJ) for A0 falling (conditional):

| Cell Name                        | When           | Power(pJ) |          |          |
|----------------------------------|----------------|-----------|----------|----------|
|                                  |                | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__aoi21_1 | (A1 * B * !Y)  | 0.01333   | 0.01338  | 0.01322  |
|                                  | (A1 * B * !Y)  | -0.00648  | -0.00648 | -0.00649 |
|                                  | (!A1 * B * !Y) | 0.01352   | 0.01388  | 0.01357  |
|                                  | (!A1 * B * !Y) | -0.00636  | -0.00648 | -0.00645 |
|                                  | (!A1 * !B * Y) | 0.01378   | 0.01388  | 0.01357  |
|                                  | (!A1 * !B * Y) | -0.00627  | -0.00645 | -0.00645 |

Passive power(pJ) for A1 rising (conditional):

| Cell Name                        | When           | Power(pJ) |          |          |
|----------------------------------|----------------|-----------|----------|----------|
|                                  |                | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__aoi21_1 | (B * !Y)       | -0.01275  | -0.01339 | -0.01325 |
|                                  | (B * !Y)       | 0.00688   | 0.00655  | 0.00652  |
|                                  | (!A0 * !B * Y) | -0.01395  | -0.01399 | -0.01410 |
|                                  | (!A0 * !B * Y) | 0.00180   | 0.00181  | 0.00173  |

Passive power(pJ) for A1 falling (conditional):

| Cell Name                        | When           | Power(pJ) |          |          |
|----------------------------------|----------------|-----------|----------|----------|
|                                  |                | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__aoi21_1 | (B * !Y)       | 0.01321   | 0.01339  | 0.01325  |
|                                  | (B * !Y)       | -0.00648  | -0.00648 | -0.00649 |
|                                  | (!A0 * !B * Y) | 0.01426   | 0.01426  | 0.01420  |
|                                  | (!A0 * !B * Y) | -0.00168  | -0.00168 | -0.00167 |

Passive power(pJ) for B rising (conditional):

| Cell Name                        | When           | Power(pJ) |          |          |
|----------------------------------|----------------|-----------|----------|----------|
|                                  |                | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__aoi21_1 | (A0 * A1 * !Y) | -0.00436  | -0.00445 | -0.00434 |
|                                  | (A0 * A1 * !Y) | 0.00781   | 0.00800  | 0.00781  |

**Passive power(pJ) for B falling (conditional):**

| Cell Name                        | When           | Power(pJ) |          |          |
|----------------------------------|----------------|-----------|----------|----------|
|                                  |                | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__aoi21_1 | (A0 * A1 * !Y) | 0.00477   | 0.00480  | 0.00453  |
|                                  | (A0 * A1 * !Y) | -0.00720  | -0.00724 | -0.00761 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_AOI22\_1

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT |    |    |    | OUTPUT |
|-------|----|----|----|--------|
| A0    | A1 | B0 | B1 | Y      |
| 0     | x  | 0  | x  | 1      |
| 0     | x  | 1  | 0  | 1      |
| x     | x  | 1  | 1  | 0      |
| 1     | 0  | 0  | x  | 1      |
| 1     | 0  | 1  | 0  | 1      |
| 1     | 1  | x  | x  | 0      |

## Footprint

| Cell Name                        | Area     |
|----------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__aoi22_1 | 34.02000 |

## Pin Capacitance Information

| Cell Name                        | Pin Cap(pf) |         |         |         | Max Cap(pf) |
|----------------------------------|-------------|---------|---------|---------|-------------|
|                                  | A0          | A1      | B0      | B1      | Y           |
| gf180mcu_osu_sc_gp9t3v3__aoi22_1 | 0.00396     | 0.00398 | 0.00404 | 0.00403 | 7.82120     |

## Leakage Information

| Cell Name                        | Leakage(nW) |         |         |
|----------------------------------|-------------|---------|---------|
|                                  | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__aoi22_1 | 0.00000     | 0.00121 | 0.00180 |

## Delay Information

Delay(ns) to Y rising :

| Cell Name                        | Timing Arc(Dir) | Delay(ns) |         |          |
|----------------------------------|-----------------|-----------|---------|----------|
|                                  |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__aoi22_1 | A0->Y (FR)      | 0.12859   | 3.26742 | 85.32590 |
|                                  | A1->Y (FR)      | 0.11098   | 3.20319 | 84.71740 |
|                                  | B0->Y (FR)      | 0.08410   | 3.83380 | 96.67700 |
|                                  | B1->Y (FR)      | 0.10054   | 3.88574 | 97.06210 |

Delay(ns) to Y falling :

| Cell Name                        | Timing Arc(Dir) | Delay(ns) |         |          |
|----------------------------------|-----------------|-----------|---------|----------|
|                                  |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__aoi22_1 | A0->Y (RF)      | 0.10602   | 2.24101 | 61.27190 |
|                                  | A1->Y (RF)      | 0.10352   | 2.81420 | 73.14220 |
|                                  | B0->Y (RF)      | 0.05493   | 2.71832 | 72.81130 |
|                                  | B1->Y (RF)      | 0.05597   | 2.14928 | 60.83530 |

## Power Information

Internal switching power(pJ) to Y rising :

| Cell Name                        | Input | Power(pJ) |         |         |
|----------------------------------|-------|-----------|---------|---------|
|                                  |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__aoi22_1 | A0    | 0.05033   | 0.21953 | 2.57131 |
|                                  | A0    | 0.00269   | 0.17119 | 2.52363 |
|                                  | A1    | 0.04138   | 0.19634 | 2.35999 |
|                                  | A1    | -0.00132  | 0.15245 | 2.31734 |
|                                  | B0    | 0.02364   | 0.17675 | 2.17919 |
|                                  | B0    | -0.00013  | 0.15277 | 2.15130 |
|                                  | B1    | 0.03206   | 0.19825 | 2.34919 |
|                                  | B1    | 0.00323   | 0.16900 | 2.31739 |

Internal switching power(pJ) to Y falling :

| Cell Name                        | Input | Power(pJ) |         |         |
|----------------------------------|-------|-----------|---------|---------|
|                                  |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__aoi22_1 | A0    | 0.01349   | 0.18312 | 2.49819 |
|                                  | A0    | 0.06099   | 0.23076 | 2.54571 |
|                                  | A1    | 0.01380   | 0.17017 | 2.20708 |
|                                  | A1    | 0.05627   | 0.21302 | 2.25172 |
|                                  | B0    | -0.00098  | 0.14866 | 2.08328 |
|                                  | B0    | 0.02280   | 0.17307 | 2.10811 |
|                                  | B1    | -0.00191  | 0.15894 | 2.33130 |
|                                  | B1    | 0.02694   | 0.18802 | 2.36014 |

Passive power(pJ) for A0 rising (conditional):

| Cell Name                        | When                 | Power(pJ) |          |          |
|----------------------------------|----------------------|-----------|----------|----------|
|                                  |                      | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__aoi22_1 | (A1 * B0 * B1 * !Y)  | -0.01256  | -0.01332 | -0.01320 |
|                                  | (A1 * B0 * B1 * !Y)  | 0.00686   | 0.00655  | 0.00652  |
|                                  | (!A1 * B0 * B1 * !Y) | -0.01352  | -0.01357 | -0.01348 |
|                                  | (!A1 * B0 * B1 * !Y) | 0.00652   | 0.00655  | 0.00642  |
|                                  | (!A1 * B0 * !B1 * Y) | -0.01352  | -0.01357 | -0.01348 |
|                                  | (!A1 * B0 * !B1 * Y) | 0.00653   | 0.00651  | 0.00642  |
|                                  | (!A1 * !B0 * Y)      | -0.01352  | -0.01357 | -0.01348 |
|                                  | (!A1 * !B0 * Y)      | 0.00653   | 0.00650  | 0.00642  |

Passive power(pJ) for A0 falling (conditional):

| Cell Name                        | When                 | Power(pJ) |          |          |
|----------------------------------|----------------------|-----------|----------|----------|
|                                  |                      | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__aoi22_1 | (A1 * B0 * B1 * !Y)  | 0.01323   | 0.01332  | 0.01320  |
|                                  | (A1 * B0 * B1 * !Y)  | -0.00647  | -0.00648 | -0.00649 |
|                                  | (!A1 * B0 * B1 * !Y) | 0.01366   | 0.01388  | 0.01357  |
|                                  | (!A1 * B0 * B1 * !Y) | -0.00636  | -0.00652 | -0.00642 |
|                                  | (!A1 * B0 * !B1 * Y) | 0.01380   | 0.01388  | 0.01357  |
|                                  | (!A1 * B0 * !B1 * Y) | -0.00624  | -0.00651 | -0.00642 |
|                                  | (!A1 * !B0 * Y)      | 0.01380   | 0.01388  | 0.01357  |
|                                  | (!A1 * !B0 * Y)      | -0.00624  | -0.00650 | -0.00642 |

Passive power(pJ) for A1 rising (conditional):

| Cell Name                        | When                   | Power(pJ) |          |          |
|----------------------------------|------------------------|-----------|----------|----------|
|                                  |                        | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__aoi22_1 | $(B0 * B1 * !Y)$       | -0.01272  | -0.01331 | -0.01317 |
|                                  | $(B0 * B1 * !Y)$       | 0.00690   | 0.00655  | 0.00652  |
|                                  | $(!A0 * B0 * !B1 * Y)$ | -0.01395  | -0.01399 | -0.01410 |
|                                  | $(!A0 * B0 * !B1 * Y)$ | 0.00180   | 0.00181  | 0.00173  |
|                                  | $(!A0 * !B0 * Y)$      | -0.01397  | -0.01399 | -0.01410 |
|                                  | $(!A0 * !B0 * Y)$      | 0.00180   | 0.00181  | 0.00173  |

Passive power(pJ) for A1 falling (conditional):

| Cell Name                        | When                   | Power(pJ) |          |          |
|----------------------------------|------------------------|-----------|----------|----------|
|                                  |                        | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__aoi22_1 | $(B0 * B1 * !Y)$       | 0.01328   | 0.01331  | 0.01317  |
|                                  | $(B0 * B1 * !Y)$       | -0.00649  | -0.00648 | -0.00648 |
|                                  | $(!A0 * B0 * !B1 * Y)$ | 0.01424   | 0.01426  | 0.01420  |
|                                  | $(!A0 * B0 * !B1 * Y)$ | -0.00167  | -0.00168 | -0.00167 |
|                                  | $(!A0 * !B0 * Y)$      | 0.01424   | 0.01426  | 0.01420  |
|                                  | $(!A0 * !B0 * Y)$      | -0.00168  | -0.00168 | -0.00167 |

Passive power(pJ) for B0 rising (conditional):

| Cell Name                        | When                   | Power(pJ) |          |          |
|----------------------------------|------------------------|-----------|----------|----------|
|                                  |                        | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__aoi22_1 | $(A0 * A1 * !Y)$       | -0.00432  | -0.00445 | -0.00434 |
|                                  | $(A0 * A1 * !Y)$       | 0.00776   | 0.00800  | 0.00781  |
|                                  | $(!A1 * !B1 * Y)$      | -0.01408  | -0.01408 | -0.01410 |
|                                  | $(!A1 * !B1 * Y)$      | 0.00182   | 0.00183  | 0.00173  |
|                                  | $(!A0 * A1 * !B1 * Y)$ | -0.01407  | -0.01408 | -0.01410 |
|                                  | $(!A0 * A1 * !B1 * Y)$ | 0.00182   | 0.00183  | 0.00173  |

Passive power(pJ) for B0 falling (conditional):



| Cell Name                        | When                 | Power(pJ) |          |          |
|----------------------------------|----------------------|-----------|----------|----------|
|                                  |                      | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__aoi22_1 | (A0 * A1 * !Y)       | 0.00493   | 0.00493  | 0.00458  |
|                                  | (A0 * A1 * !Y)       | -0.00706  | -0.00706 | -0.00755 |
|                                  | (!A1 * !B1 * Y)      | 0.01426   | 0.01425  | 0.01420  |
|                                  | (!A1 * !B1 * Y)      | -0.00172  | -0.00168 | -0.00167 |
|                                  | (!A0 * A1 * !B1 * Y) | 0.01426   | 0.01425  | 0.01420  |
|                                  | (!A0 * A1 * !B1 * Y) | -0.00171  | -0.00168 | -0.00167 |

Passive power(pJ) for B1 rising (conditional):

| Cell Name                        | When                 | Power(pJ) |          |          |
|----------------------------------|----------------------|-----------|----------|----------|
|                                  |                      | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__aoi22_1 | (A0 * A1 * !Y)       | -0.00437  | -0.00444 | -0.00434 |
|                                  | (A0 * A1 * !Y)       | 0.00783   | 0.00800  | 0.00781  |
|                                  | (!A1 * !B0 * Y)      | -0.01351  | -0.01353 | -0.01352 |
|                                  | (!A1 * !B0 * Y)      | 0.00645   | 0.00649  | 0.00644  |
|                                  | (!A0 * A1 * !B0 * Y) | -0.01355  | -0.01359 | -0.01348 |
|                                  | (!A0 * A1 * !B0 * Y) | 0.00652   | 0.00659  | 0.00643  |

Passive power(pJ) for B1 falling (conditional):

| Cell Name                        | When                 | Power(pJ) |          |          |
|----------------------------------|----------------------|-----------|----------|----------|
|                                  |                      | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__aoi22_1 | (A0 * A1 * !Y)       | 0.00491   | 0.00493  | 0.00458  |
|                                  | (A0 * A1 * !Y)       | -0.00702  | -0.00706 | -0.00755 |
|                                  | (!A1 * !B0 * Y)      | 0.01359   | 0.01360  | 0.01357  |
|                                  | (!A1 * !B0 * Y)      | -0.00631  | -0.00649 | -0.00644 |
|                                  | (!A0 * A1 * !B0 * Y) | 0.01360   | 0.01360  | 0.01357  |
|                                  | (!A0 * A1 * !B0 * Y) | -0.00630  | -0.00649 | -0.00643 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_BUF\_16

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT | OUTPUT |
|-------|--------|
| A     | Y      |
| 0     | 0      |
| 1     | 1      |

## Footprint

| Cell Name                       | Area      |
|---------------------------------|-----------|
| gf180mcu_osu_sc_gp9t3v3__buf_16 | 100.17000 |

## Pin Capacitance Information

| Cell Name                       | Pin Cap(pf) | Max Cap(pf) |
|---------------------------------|-------------|-------------|
|                                 | A           | Y           |
| gf180mcu_osu_sc_gp9t3v3__buf_16 | 0.00404     | 248.43445   |

## Leakage Information

| Cell Name                       | Leakage(nW) |         |         |
|---------------------------------|-------------|---------|---------|
|                                 | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__buf_16 | 0.00000     | 0.01253 | 0.01497 |

## Delay Information

Delay(ns) to Y rising :

| Cell Name                       | Timing Arc(Dir) | Delay(ns) |         |          |
|---------------------------------|-----------------|-----------|---------|----------|
|                                 |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__buf_16 | A->Y (RR)       | 0.33095   | 1.69045 | 68.99860 |

Delay(ns) to Y falling :

| Cell Name                       | Timing Arc(Dir) | Delay(ns) |         |          |
|---------------------------------|-----------------|-----------|---------|----------|
|                                 |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__buf_16 | A->Y (FF)       | 0.35633   | 2.36687 | 75.57860 |

## Power Information

Internal switching power(pJ) to Y rising :

| Cell Name                       | Input | Power(pJ) |         |         |
|---------------------------------|-------|-----------|---------|---------|
|                                 |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__buf_16 | A     | 0.71814   | 1.10493 | 7.51130 |
|                                 | A     | 0.74011   | 1.12616 | 7.53316 |

Internal switching power(pJ) to Y falling :

| Cell Name                       | Input | Power(pJ) |         |         |
|---------------------------------|-------|-----------|---------|---------|
|                                 |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__buf_16 | A     | 0.77223   | 1.11841 | 7.44280 |
|                                 | A     | 0.75017   | 1.09709 | 7.47522 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_BUF\_1

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT | OUTPUT |
|-------|--------|
| A     | Y      |
| 0     | 0      |
| 1     | 1      |

## Footprint

| Cell Name                      | Area     |
|--------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__buf_1 | 20.16000 |

## Pin Capacitance Information

| Cell Name                      | Pin Cap(pf) | Max Cap(pf) |
|--------------------------------|-------------|-------------|
|                                | A           | Y           |
| gf180mcu_osu_sc_gp9t3v3__buf_1 | 0.00405     | 15.60333    |

## Leakage Information

| Cell Name                      | Leakage(nW) |         |         |
|--------------------------------|-------------|---------|---------|
|                                | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__buf_1 | 0.00000     | 0.00147 | 0.00147 |

## Delay Information

Delay(ns) to Y rising :

| Cell Name                      | Timing Arc(Dir) | Delay(ns) |         |          |
|--------------------------------|-----------------|-----------|---------|----------|
|                                |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__buf_1 | A->Y (RR)       | 0.07224   | 1.71490 | 67.36110 |

Delay(ns) to Y falling :

| Cell Name                      | Timing Arc(Dir) | Delay(ns) |         |          |
|--------------------------------|-----------------|-----------|---------|----------|
|                                |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__buf_1 | A->Y (FF)       | 0.08009   | 2.29427 | 73.88930 |

## Power Information

Internal switching power(pJ) to Y rising :

| Cell Name                      | Input | Power(pJ) |         |         |
|--------------------------------|-------|-----------|---------|---------|
|                                |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__buf_1 | A     | 0.01394   | 0.38726 | 6.93368 |
|                                | A     | 0.03581   | 0.40900 | 6.95543 |

Internal switching power(pJ) to Y falling :

| Cell Name                      | Input | Power(pJ) |         |         |
|--------------------------------|-------|-----------|---------|---------|
|                                |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__buf_1 | A     | 0.03604   | 0.41170 | 6.94970 |
|                                | A     | 0.01419   | 0.38995 | 6.92784 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_BUF\_2

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT | OUTPUT |
|-------|--------|
| A     | Y      |
| 0     | 0      |
| 1     | 1      |

## Footprint

| Cell Name                      | Area     |
|--------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__buf_2 | 24.57000 |

## Pin Capacitance Information

| Cell Name                      | Pin Cap(pf) | Max Cap(pf) |
|--------------------------------|-------------|-------------|
|                                | A           | Y           |
| gf180mcu_osu_sc_gp9t3v3__buf_2 | 0.00405     | 31.24748    |

## Leakage Information

| Cell Name                      | Leakage(nW) |         |         |
|--------------------------------|-------------|---------|---------|
|                                | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__buf_2 | 0.00000     | 0.00221 | 0.00237 |



## Delay Information

Delay(ns) to Y rising :

| Cell Name                      | Timing Arc(Dir) | Delay(ns) |         |          |
|--------------------------------|-----------------|-----------|---------|----------|
|                                |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__buf_2 | A->Y (RR)       | 0.08733   | 1.40493 | 67.60510 |

Delay(ns) to Y falling :

| Cell Name                      | Timing Arc(Dir) | Delay(ns) |         |          |
|--------------------------------|-----------------|-----------|---------|----------|
|                                |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__buf_2 | A->Y (FF)       | 0.09581   | 2.02505 | 74.13340 |

## Power Information

Internal switching power(pJ) to Y rising :

| Cell Name                      | Input | Power(pJ) |         |         |
|--------------------------------|-------|-----------|---------|---------|
|                                |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__buf_2 | A     | 0.03373   | 0.40878 | 6.95726 |
|                                | A     | 0.05567   | 0.43051 | 6.97903 |

Internal switching power(pJ) to Y falling :

| Cell Name                      | Input | Power(pJ) |         |         |
|--------------------------------|-------|-----------|---------|---------|
|                                |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__buf_2 | A     | 0.05414   | 0.43050 | 6.97356 |
|                                | A     | 0.03215   | 0.40907 | 6.96379 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_BUF\_4

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT | OUTPUT |
|-------|--------|
| A     | Y      |
| 0     | 0      |
| 1     | 1      |

## Footprint

| Cell Name                      | Area     |
|--------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__buf_4 | 35.91000 |

## Pin Capacitance Information

| Cell Name                      | Pin Cap(pf) | Max Cap(pf) |
|--------------------------------|-------------|-------------|
|                                | A           | Y           |
| gf180mcu_osu_sc_gp9t3v3__buf_4 | 0.00404     | 62.37794    |

## Leakage Information

| Cell Name                      | Leakage(nW) |         |         |
|--------------------------------|-------------|---------|---------|
|                                | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__buf_4 | 0.00000     | 0.00369 | 0.00417 |

## Delay Information

Delay(ns) to Y rising :

| Cell Name                      | Timing Arc(Dir) | Delay(ns) |         |          |
|--------------------------------|-----------------|-----------|---------|----------|
|                                |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__buf_4 | A->Y (RR)       | 0.12024   | 1.25211 | 67.74940 |

Delay(ns) to Y falling :

| Cell Name                      | Timing Arc(Dir) | Delay(ns) |         |          |
|--------------------------------|-----------------|-----------|---------|----------|
|                                |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__buf_4 | A->Y (FF)       | 0.13068   | 1.90125 | 74.31160 |

## Power Information

Internal switching power(pJ) to Y rising :

| Cell Name                      | Input | Power(pJ) |         |         |
|--------------------------------|-------|-----------|---------|---------|
|                                |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__buf_4 | A     | 0.08282   | 0.46664 | 7.00894 |
|                                | A     | 0.10486   | 0.48809 | 7.03028 |

Internal switching power(pJ) to Y falling :

| Cell Name                      | Input | Power(pJ) |         |         |
|--------------------------------|-------|-----------|---------|---------|
|                                |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__buf_4 | A     | 0.10190   | 0.48425 | 7.02004 |
|                                | A     | 0.07977   | 0.46257 | 7.02242 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_BUF\_8

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT | OUTPUT |
|-------|--------|
| A     | Y      |
| 0     | 0      |
| 1     | 1      |

## Footprint

| Cell Name                      | Area     |
|--------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__buf_8 | 58.59000 |

## Pin Capacitance Information

| Cell Name                      | Pin Cap(pf) | Max Cap(pf) |
|--------------------------------|-------------|-------------|
|                                | A           | Y           |
| gf180mcu_osu_sc_gp9t3v3__buf_8 | 0.00404     | 123.63758   |

## Leakage Information

| Cell Name                      | Leakage(nW) |         |         |
|--------------------------------|-------------|---------|---------|
|                                | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__buf_8 | 0.00000     | 0.00663 | 0.00777 |

## Delay Information

Delay(ns) to Y rising :

| Cell Name                      | Timing Arc(Dir) | Delay(ns) |         |          |
|--------------------------------|-----------------|-----------|---------|----------|
|                                |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__buf_8 | A->Y (RR)       | 0.18674   | 1.27334 | 67.65870 |

Delay(ns) to Y falling :

| Cell Name                      | Timing Arc(Dir) | Delay(ns) |         |          |
|--------------------------------|-----------------|-----------|---------|----------|
|                                |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__buf_8 | A->Y (FF)       | 0.20220   | 1.95122 | 74.29590 |

## Power Information

Internal switching power(pJ) to Y rising :

| Cell Name                      | Input | Power(pJ) |         |         |
|--------------------------------|-------|-----------|---------|---------|
|                                |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__buf_8 | A     | 0.22690   | 0.63680 | 7.31047 |
|                                | A     | 0.24885   | 0.65843 | 7.17867 |

Internal switching power(pJ) to Y falling :

| Cell Name                      | Input | Power(pJ) |         |         |
|--------------------------------|-------|-----------|---------|---------|
|                                |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__buf_8 | A     | 0.24824   | 0.64435 | 7.13433 |
|                                | A     | 0.22625   | 0.62247 | 7.16156 |



# GF180MCU\_OSU\_SC\_GP9T3V3\_\_CLKBUF\_16

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT | OUTPUT |
|-------|--------|
| A     | Y      |
| 0     | 0      |
| 1     | 1      |

## Footprint

| Cell Name                          | Area      |
|------------------------------------|-----------|
| gf180mcu_osu_sc_gp9t3v3__clkbuf_16 | 100.17000 |

## Pin Capacitance Information

| Cell Name                          | Pin Cap(pf) | Max Cap(pf) |
|------------------------------------|-------------|-------------|
|                                    | A           | Y           |
| gf180mcu_osu_sc_gp9t3v3__clkbuf_16 | 0.00404     | 248.43445   |

## Leakage Information

| Cell Name                          | Leakage(nW) |         |         |
|------------------------------------|-------------|---------|---------|
|                                    | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__clkbuf_16 | 0.00000     | 0.01253 | 0.01497 |

## Delay Information

Delay(ns) to Y rising :

| Cell Name                          | Timing Arc(Dir) | Delay(ns) |         |          |
|------------------------------------|-----------------|-----------|---------|----------|
|                                    |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__clkbuf_16 | A->Y (RR)       | 0.33095   | 1.69045 | 68.99860 |

Delay(ns) to Y falling :

| Cell Name                          | Timing Arc(Dir) | Delay(ns) |         |          |
|------------------------------------|-----------------|-----------|---------|----------|
|                                    |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__clkbuf_16 | A->Y (FF)       | 0.35633   | 2.36687 | 75.57860 |

## Power Information

Internal switching power(pJ) to Y rising :

| Cell Name                          | Input | Power(pJ) |         |         |
|------------------------------------|-------|-----------|---------|---------|
|                                    |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__clkbuf_16 | A     | 0.71814   | 1.10493 | 7.51130 |
|                                    | A     | 0.74011   | 1.12616 | 7.53316 |

Internal switching power(pJ) to Y falling :

| Cell Name                          | Input | Power(pJ) |         |         |
|------------------------------------|-------|-----------|---------|---------|
|                                    |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__clkbuf_16 | A     | 0.77223   | 1.11841 | 7.44280 |
|                                    | A     | 0.75017   | 1.09709 | 7.47522 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_CLKBUF\_1

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT | OUTPUT |
|-------|--------|
| A     | Y      |
| 0     | 0      |
| 1     | 1      |

## Footprint

| Cell Name                         | Area     |
|-----------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__clkbuf_1 | 20.79000 |

## Pin Capacitance Information

| Cell Name                         | Pin Cap(pf) | Max Cap(pf) |
|-----------------------------------|-------------|-------------|
|                                   | A           | Y           |
| gf180mcu_osu_sc_gp9t3v3__clkbuf_1 | 0.00405     | 15.60333    |

## Leakage Information

| Cell Name                         | Leakage(nW) |         |         |
|-----------------------------------|-------------|---------|---------|
|                                   | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__clkbuf_1 | 0.00000     | 0.00147 | 0.00147 |

## Delay Information

Delay(ns) to Y rising :

| Cell Name                         | Timing Arc(Dir) | Delay(ns) |         |          |
|-----------------------------------|-----------------|-----------|---------|----------|
|                                   |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__clkbuf_1 | A->Y (RR)       | 0.07224   | 1.71490 | 67.36110 |

Delay(ns) to Y falling :

| Cell Name                         | Timing Arc(Dir) | Delay(ns) |         |          |
|-----------------------------------|-----------------|-----------|---------|----------|
|                                   |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__clkbuf_1 | A->Y (FF)       | 0.08009   | 2.29427 | 73.88930 |

## Power Information

Internal switching power(pJ) to Y rising :

| Cell Name                         | Input | Power(pJ) |         |         |
|-----------------------------------|-------|-----------|---------|---------|
|                                   |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__clkbuf_1 | A     | 0.01394   | 0.38726 | 6.93368 |
|                                   | A     | 0.03581   | 0.40900 | 6.95543 |

Internal switching power(pJ) to Y falling :

| Cell Name                         | Input | Power(pJ) |         |         |
|-----------------------------------|-------|-----------|---------|---------|
|                                   |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__clkbuf_1 | A     | 0.03604   | 0.41170 | 6.94970 |
|                                   | A     | 0.01419   | 0.38995 | 6.92784 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_CLKBUF\_2

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT | OUTPUT |
|-------|--------|
| A     | Y      |
| 0     | 0      |
| 1     | 1      |

## Footprint

| Cell Name                         | Area     |
|-----------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__clkbuf_2 | 24.57000 |

## Pin Capacitance Information

| Cell Name                         | Pin Cap(pf) | Max Cap(pf) |
|-----------------------------------|-------------|-------------|
|                                   | A           | Y           |
| gf180mcu_osu_sc_gp9t3v3__clkbuf_2 | 0.00405     | 31.24748    |

## Leakage Information

| Cell Name                         | Leakage(nW) |         |         |
|-----------------------------------|-------------|---------|---------|
|                                   | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__clkbuf_2 | 0.00000     | 0.00221 | 0.00237 |

## Delay Information

Delay(ns) to Y rising :

| Cell Name                         | Timing Arc(Dir) | Delay(ns) |         |          |
|-----------------------------------|-----------------|-----------|---------|----------|
|                                   |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__clkbuf_2 | A->Y (RR)       | 0.08733   | 1.40493 | 67.60510 |

Delay(ns) to Y falling :

| Cell Name                         | Timing Arc(Dir) | Delay(ns) |         |          |
|-----------------------------------|-----------------|-----------|---------|----------|
|                                   |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__clkbuf_2 | A->Y (FF)       | 0.09581   | 2.02505 | 74.13340 |



## Power Information

Internal switching power(pJ) to Y rising :

| Cell Name                         | Input | Power(pJ) |         |         |
|-----------------------------------|-------|-----------|---------|---------|
|                                   |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__clkbuf_2 | A     | 0.03373   | 0.40878 | 6.95726 |
|                                   | A     | 0.05567   | 0.43051 | 6.97903 |

Internal switching power(pJ) to Y falling :

| Cell Name                         | Input | Power(pJ) |         |         |
|-----------------------------------|-------|-----------|---------|---------|
|                                   |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__clkbuf_2 | A     | 0.05414   | 0.43050 | 6.97356 |
|                                   | A     | 0.03215   | 0.40907 | 6.96379 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_CLKBUF\_4

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT | OUTPUT |
|-------|--------|
| A     | Y      |
| 0     | 0      |
| 1     | 1      |

## Footprint

| Cell Name                         | Area     |
|-----------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__clkbuf_4 | 35.91000 |

## Pin Capacitance Information

| Cell Name                         | Pin Cap(pf) | Max Cap(pf) |
|-----------------------------------|-------------|-------------|
|                                   | A           | Y           |
| gf180mcu_osu_sc_gp9t3v3__clkbuf_4 | 0.00404     | 62.37794    |

## Leakage Information

| Cell Name                         | Leakage(nW) |         |         |
|-----------------------------------|-------------|---------|---------|
|                                   | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__clkbuf_4 | 0.00000     | 0.00369 | 0.00417 |

## Delay Information

Delay(ns) to Y rising :

| Cell Name                         | Timing Arc(Dir) | Delay(ns) |         |          |
|-----------------------------------|-----------------|-----------|---------|----------|
|                                   |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__clkbuf_4 | A->Y (RR)       | 0.12024   | 1.25211 | 67.74940 |

Delay(ns) to Y falling :

| Cell Name                         | Timing Arc(Dir) | Delay(ns) |         |          |
|-----------------------------------|-----------------|-----------|---------|----------|
|                                   |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__clkbuf_4 | A->Y (FF)       | 0.13068   | 1.90125 | 74.31160 |

## Power Information

Internal switching power(pJ) to Y rising :

| Cell Name                         | Input | Power(pJ) |         |         |
|-----------------------------------|-------|-----------|---------|---------|
|                                   |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__clkbuf_4 | A     | 0.08282   | 0.46664 | 7.00894 |
|                                   | A     | 0.10486   | 0.48809 | 7.03028 |

Internal switching power(pJ) to Y falling :

| Cell Name                         | Input | Power(pJ) |         |         |
|-----------------------------------|-------|-----------|---------|---------|
|                                   |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__clkbuf_4 | A     | 0.10190   | 0.48425 | 7.02004 |
|                                   | A     | 0.07977   | 0.46257 | 7.02242 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_CLKBUF\_8

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT | OUTPUT |
|-------|--------|
| A     | Y      |
| 0     | 0      |
| 1     | 1      |

## Footprint

| Cell Name                         | Area     |
|-----------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__clkbuf_8 | 58.59000 |

## Pin Capacitance Information

| Cell Name                         | Pin Cap(pf) | Max Cap(pf) |
|-----------------------------------|-------------|-------------|
|                                   | A           | Y           |
| gf180mcu_osu_sc_gp9t3v3__clkbuf_8 | 0.00404     | 123.63758   |

## Leakage Information

| Cell Name                         | Leakage(nW) |         |         |
|-----------------------------------|-------------|---------|---------|
|                                   | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__clkbuf_8 | 0.00000     | 0.00663 | 0.00777 |

## Delay Information

Delay(ns) to Y rising :

| Cell Name                         | Timing Arc(Dir) | Delay(ns) |         |          |
|-----------------------------------|-----------------|-----------|---------|----------|
|                                   |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__clkbuf_8 | A->Y (RR)       | 0.18674   | 1.27334 | 67.65870 |

Delay(ns) to Y falling :

| Cell Name                         | Timing Arc(Dir) | Delay(ns) |         |          |
|-----------------------------------|-----------------|-----------|---------|----------|
|                                   |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__clkbuf_8 | A->Y (FF)       | 0.20220   | 1.95122 | 74.29590 |

## Power Information

Internal switching power(pJ) to Y rising :

| Cell Name                         | Input | Power(pJ) |         |         |
|-----------------------------------|-------|-----------|---------|---------|
|                                   |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__clkbuf_8 | A     | 0.22690   | 0.63680 | 7.31047 |
|                                   | A     | 0.24885   | 0.65843 | 7.17867 |

Internal switching power(pJ) to Y falling :

| Cell Name                         | Input | Power(pJ) |         |         |
|-----------------------------------|-------|-----------|---------|---------|
|                                   |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__clkbuf_8 | A     | 0.24824   | 0.64435 | 7.13433 |
|                                   | A     | 0.22625   | 0.62247 | 7.16156 |

## GF180MCU\_OSU\_SC\_GP9T3V3\_\_CLKINV\_16

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ecs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

### Truth Table

| INPUT | OUTPUT |
|-------|--------|
| A     | Y      |
| 0     | 1      |
| 1     | 0      |

### Footprint

| Cell Name                          | Area     |
|------------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__clkinv_16 | 94.50000 |

### Pin Capacitance Information

| Cell Name                          | Pin Cap(pf) | Max Cap(pf) |
|------------------------------------|-------------|-------------|
|                                    | A           | Y           |
| gf180mcu_osu_sc_gp9t3v3__clkinv_16 | 0.06475     | 234.89906   |

### Leakage Information

| Cell Name                          | Leakage(nW) |         |         |
|------------------------------------|-------------|---------|---------|
|                                    | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__clkinv_16 | 0.00000     | 0.01179 | 0.01439 |



## Delay Information

Delay(ns) to Y rising :

| Cell Name                          | Timing Arc(Dir) | Delay(ns) |         |          |
|------------------------------------|-----------------|-----------|---------|----------|
|                                    |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__clkinv_16 | A->Y (FR)       | 0.03119   | 2.15006 | 98.41110 |

Delay(ns) to Y falling :

| Cell Name                          | Timing Arc(Dir) | Delay(ns) |         |          |
|------------------------------------|-----------------|-----------|---------|----------|
|                                    |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__clkinv_16 | A->Y (RF)       | 0.02336   | 1.37517 | 83.67180 |

## Power Information

Internal switching power(pJ) to Y rising :

| Cell Name                          | Input | Power(pJ) |         |          |
|------------------------------------|-------|-----------|---------|----------|
|                                    |       | first     | mid     | last     |
| gf180mcu_osu_sc_gp9t3v3__clkinv_16 | A     | 0.32278   | 4.42993 | 38.24140 |
|                                    | A     | -0.02776  | 4.07468 | 37.86540 |

Internal switching power(pJ) to Y falling :

| Cell Name                          | Input | Power(pJ) |         |          |
|------------------------------------|-------|-----------|---------|----------|
|                                    |       | first     | mid     | last     |
| gf180mcu_osu_sc_gp9t3v3__clkinv_16 | A     | -0.05399  | 3.89215 | 34.47270 |
|                                    | A     | 0.29632   | 4.24422 | 34.82460 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_CLKINV\_1

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT | OUTPUT |
|-------|--------|
| A     | Y      |
| 0     | 1      |
| 1     | 0      |

## Footprint

| Cell Name                         | Area     |
|-----------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__clkinv_1 | 15.12000 |

## Pin Capacitance Information

| Cell Name                         | Pin Cap(pf) | Max Cap(pf) |
|-----------------------------------|-------------|-------------|
|                                   | A           | Y           |
| gf180mcu_osu_sc_gp9t3v3__clkinv_1 | 0.00405     | 14.68244    |

## Leakage Information

| Cell Name                         | Leakage(nW) |         |         |
|-----------------------------------|-------------|---------|---------|
|                                   | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__clkinv_1 | 0.00000     | 0.00074 | 0.00090 |

## Delay Information

Delay(ns) to Y rising :

| Cell Name                         | Timing Arc(Dir) | Delay(ns) |         |          |
|-----------------------------------|-----------------|-----------|---------|----------|
|                                   |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__clkinv_1 | A->Y (FR)       | 0.03869   | 3.38170 | 98.41570 |

Delay(ns) to Y falling :

| Cell Name                         | Timing Arc(Dir) | Delay(ns) |         |          |
|-----------------------------------|-----------------|-----------|---------|----------|
|                                   |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__clkinv_1 | A->Y (RF)       | 0.03096   | 2.61905 | 83.67590 |

## Power Information

Internal switching power(pJ) to Y rising :

| Cell Name                         | Input | Power(pJ) |         |         |
|-----------------------------------|-------|-----------|---------|---------|
|                                   |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__clkinv_1 | A     | 0.01992   | 0.22335 | 2.39002 |
|                                   | A     | -0.00196  | 0.20134 | 2.36651 |

Internal switching power(pJ) to Y falling :

| Cell Name                         | Input | Power(pJ) |         |         |
|-----------------------------------|-------|-----------|---------|---------|
|                                   |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__clkinv_1 | A     | -0.00449  | 0.17814 | 2.15348 |
|                                   | A     | 0.01738   | 0.20005 | 2.17547 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_CLKINV\_2

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT | OUTPUT |
|-------|--------|
| A     | Y      |
| 0     | 1      |
| 1     | 0      |

## Footprint

| Cell Name                         | Area     |
|-----------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__clkinv_2 | 20.79000 |

## Pin Capacitance Information

| Cell Name                         | Pin Cap(pf) | Max Cap(pf) |
|-----------------------------------|-------------|-------------|
|                                   | A           | Y           |
| gf180mcu_osu_sc_gp9t3v3__clkinv_2 | 0.00808     | 29.36374    |

## Leakage Information

| Cell Name                         | Leakage(nW) |         |         |
|-----------------------------------|-------------|---------|---------|
|                                   | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__clkinv_2 | 0.00000     | 0.00147 | 0.00180 |

## Delay Information

Delay(ns) to Y rising :

| Cell Name                         | Timing Arc(Dir) | Delay(ns) |         |          |
|-----------------------------------|-----------------|-----------|---------|----------|
|                                   |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__clkinv_2 | A->Y (FR)       | 0.03414   | 2.91144 | 98.41320 |

Delay(ns) to Y falling :

| Cell Name                         | Timing Arc(Dir) | Delay(ns) |         |          |
|-----------------------------------|-----------------|-----------|---------|----------|
|                                   |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__clkinv_2 | A->Y (RF)       | 0.02634   | 2.15085 | 83.67370 |

## Power Information

Internal switching power(pJ) to Y rising :

| Cell Name                         | Input | Power(pJ) |         |         |
|-----------------------------------|-------|-----------|---------|---------|
|                                   |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__clkinv_2 | A     | 0.03999   | 0.47013 | 4.78011 |
|                                   | A     | -0.00363  | 0.42539 | 4.73310 |

Internal switching power(pJ) to Y falling :

| Cell Name                         | Input | Power(pJ) |         |         |
|-----------------------------------|-------|-----------|---------|---------|
|                                   |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__clkinv_2 | A     | -0.00881  | 0.39669 | 4.30699 |
|                                   | A     | 0.03480   | 0.44042 | 4.35098 |



# GF180MCU\_OSU\_SC\_GP9T3V3\_\_CLKINV\_4

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT | OUTPUT |
|-------|--------|
| A     | Y      |
| 0     | 1      |
| 1     | 0      |

## Footprint

| Cell Name                         | Area     |
|-----------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__clkinv_4 | 30.24000 |

## Pin Capacitance Information

| Cell Name                         | Pin Cap(pf) | Max Cap(pf) |
|-----------------------------------|-------------|-------------|
|                                   | A           | Y           |
| gf180mcu_osu_sc_gp9t3v3__clkinv_4 | 0.01618     | 58.72635    |

## Leakage Information

| Cell Name                         | Leakage(nW) |         |         |
|-----------------------------------|-------------|---------|---------|
|                                   | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__clkinv_4 | 0.00000     | 0.00295 | 0.00360 |

## Delay Information

Delay(ns) to Y rising :

| Cell Name                         | Timing Arc(Dir) | Delay(ns) |         |          |
|-----------------------------------|-----------------|-----------|---------|----------|
|                                   |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__clkinv_4 | A->Y (FR)       | 0.03168   | 2.52828 | 98.41200 |

Delay(ns) to Y falling :

| Cell Name                         | Timing Arc(Dir) | Delay(ns) |         |          |
|-----------------------------------|-----------------|-----------|---------|----------|
|                                   |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__clkinv_4 | A->Y (RF)       | 0.02386   | 1.76026 | 83.67260 |

## Power Information

Internal switching power(pJ) to Y rising :

| Cell Name                         | Input | Power(pJ) |         |         |
|-----------------------------------|-------|-----------|---------|---------|
|                                   |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__clkinv_4 | A     | 0.08065   | 1.02183 | 9.56029 |
|                                   | A     | -0.00700  | 0.93247 | 9.46628 |

Internal switching power(pJ) to Y falling :

| Cell Name                         | Input | Power(pJ) |         |         |
|-----------------------------------|-------|-----------|---------|---------|
|                                   |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__clkinv_4 | A     | -0.01761  | 0.87513 | 8.61403 |
|                                   | A     | 0.06989   | 0.96373 | 8.70200 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_CLKINV\_8

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT | OUTPUT |
|-------|--------|
| A     | Y      |
| 0     | 1      |
| 1     | 0      |

## Footprint

| Cell Name                         | Area     |
|-----------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__clkinv_8 | 52.92000 |

## Pin Capacitance Information

| Cell Name                         | Pin Cap(pf) | Max Cap(pf) |
|-----------------------------------|-------------|-------------|
|                                   | A           | Y           |
| gf180mcu_osu_sc_gp9t3v3__clkinv_8 | 0.03246     | 117.45156   |

## Leakage Information

| Cell Name                         | Leakage(nW) |         |         |
|-----------------------------------|-------------|---------|---------|
|                                   | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__clkinv_8 | 0.00000     | 0.00590 | 0.00720 |

## Delay Information

Delay(ns) to Y rising :

| Cell Name                         | Timing Arc(Dir) | Delay(ns) |         |          |
|-----------------------------------|-----------------|-----------|---------|----------|
|                                   |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__clkinv_8 | A->Y (FR)       | 0.03034   | 2.20462 | 98.41140 |

Delay(ns) to Y falling :

| Cell Name                         | Timing Arc(Dir) | Delay(ns) |         |          |
|-----------------------------------|-----------------|-----------|---------|----------|
|                                   |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__clkinv_8 | A->Y (RF)       | 0.02253   | 1.43037 | 83.67200 |

## Power Information

Internal switching power(pJ) to Y rising :

| Cell Name                         | Input | Power(pJ) |         |          |
|-----------------------------------|-------|-----------|---------|----------|
|                                   |       | first     | mid     | last     |
| gf180mcu_osu_sc_gp9t3v3__clkinv_8 | A     | 0.16151   | 2.18866 | 19.12070 |
|                                   | A     | -0.01369  | 2.01180 | 18.93270 |

Internal switching power(pJ) to Y falling :

| Cell Name                         | Input | Power(pJ) |         |          |
|-----------------------------------|-------|-----------|---------|----------|
|                                   |       | first     | mid     | last     |
| gf180mcu_osu_sc_gp9t3v3__clkinv_8 | A     | -0.03509  | 1.90803 | 17.22810 |
|                                   | A     | 0.14004   | 2.08375 | 17.40400 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_DECAP\_1

*gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs*  
*Cell Library: Process , Voltage 3.30,*  
*Temp 25.00*

---

## Footprint

| Cell Name                        | Area     |
|----------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__decap_1 | 15.12000 |

## Pin Capacitance Information

## Leakage Information

| Cell Name                        | Leakage(nW) |         |         |
|----------------------------------|-------------|---------|---------|
|                                  | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__decap_1 | 0.00000     | 0.00000 | 0.00000 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_DFFN\_1

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT |     | OUTPUT |     |
|-------|-----|--------|-----|
| D     | CLK | Q      | QN  |
| 0     | F   | 0      | 1   |
| 1     | F   | 1      | 0   |
| x     | x   | IQ     | IQN |

## Footprint

| Cell Name                       | Area     |
|---------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__dffn_1 | 98.28000 |

## Pin Capacitance Information

| Cell Name                       | Pin Cap(pf) |         | Max Cap(pf) |          |
|---------------------------------|-------------|---------|-------------|----------|
|                                 | D           | CLK     | Q           | QN       |
| gf180mcu_osu_sc_gp9t3v3__dffn_1 | 0.00395     | 0.00407 | 15.56757    | 15.73192 |

## Leakage Information

| Cell Name                       | Leakage(nW) |              |              |
|---------------------------------|-------------|--------------|--------------|
|                                 | Min.        | Avg          | Max.         |
| gf180mcu_osu_sc_gp9t3v3__dffn_1 | 0.00000     | 100780.00000 | 174312.00000 |



## Delay Information

Delay(ns) to Q rising :

| Cell Name                       | Timing Arc(Dir) | Delay(ns) |         |           |
|---------------------------------|-----------------|-----------|---------|-----------|
|                                 |                 | First     | Mid     | Last      |
| gf180mcu_osu_sc_gp9t3v3__dffn_1 | CLK->Q (FR)     | 0.34664   | 4.73230 | 169.77500 |
|                                 | QN->Q (FR)      | 0.03869   | 3.44899 | 102.18800 |

Delay(ns) to Q falling :

| Cell Name                       | Timing Arc(Dir) | Delay(ns) |         |           |
|---------------------------------|-----------------|-----------|---------|-----------|
|                                 |                 | First     | Mid     | Last      |
| gf180mcu_osu_sc_gp9t3v3__dffn_1 | CLK->Q (FF)     | 0.63732   | 5.05970 | 169.08500 |
|                                 | QN->Q (RF)      | 0.03096   | 2.68465 | 87.05470  |

Delay(ns) to QN rising :

| Cell Name                       | Timing Arc(Dir) | Delay(ns) |         |          |
|---------------------------------|-----------------|-----------|---------|----------|
|                                 |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__dffn_1 | CLK->QN (FR)    | 0.60938   | 2.85804 | 77.38850 |

Delay(ns) to QN falling :

| Cell Name                       | Timing Arc(Dir) | Delay(ns) |         |          |
|---------------------------------|-----------------|-----------|---------|----------|
|                                 |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__dffn_1 | CLK->QN (FF)    | 0.31360   | 2.28826 | 67.87030 |

## Constraint Information

Constraints(ns) for D rising :

| Cell Name                       | Timing Check | Ref Pin(trans) | Reference Slew Rate(ns) |         |          |
|---------------------------------|--------------|----------------|-------------------------|---------|----------|
|                                 |              |                | first                   | mid     | last     |
| gf180mcu_osu_sc_gp9t3v3__dffn_1 | hold         | CLK (F)        | 0.01420                 | 0.11279 | -3.15281 |
|                                 | setup        | CLK (F)        | 0.32167                 | 0.85563 | 29.09670 |

Constraints(ns) for D falling :

| Cell Name                       | Timing Check | Ref Pin(trans) | Reference Slew Rate(ns) |          |           |
|---------------------------------|--------------|----------------|-------------------------|----------|-----------|
|                                 |              |                | first                   | mid      | last      |
| gf180mcu_osu_sc_gp9t3v3__dffn_1 | hold         | CLK (F)        | -0.30194                | -1.79930 | 46.00920  |
|                                 | setup        | CLK (F)        | 0.44366                 | 1.80271  | -31.50730 |

Constraints(ns) for CLK rising (conditional):

| Cell Name                       | Timing Check    | Ref Pin(trans) | When | Reference Slew Rate(ns) |         |           |
|---------------------------------|-----------------|----------------|------|-------------------------|---------|-----------|
|                                 |                 |                |      | first                   | mid     | last      |
| gf180mcu_osu_sc_gp9t3v3__dffn_1 | min_pulse_width | CLK ()         | D    | 0.38545                 | 4.36523 | 287.64800 |
|                                 | min_pulse_width | CLK ()         | !D   | 0.36637                 | 4.36523 | 287.26500 |

Constraints(ns) for CLK falling (conditional):

| Cell Name                       | Timing Check    | Ref Pin(trans) | When | Reference Slew Rate(ns) |         |           |
|---------------------------------|-----------------|----------------|------|-------------------------|---------|-----------|
|                                 |                 |                |      | first                   | mid     | last      |
| gf180mcu_osu_sc_gp9t3v3__dffn_1 | min_pulse_width | CLK ()         | D    | 0.21851                 | 4.36523 | 165.00100 |
|                                 | min_pulse_width | CLK ()         | !D   | 0.14220                 | 4.36523 | 165.00100 |

## Power Information

Internal switching power(pJ) to Q rising :

| Cell Name                       | Input | Power(pJ) |         |         |
|---------------------------------|-------|-----------|---------|---------|
|                                 |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__dffn_1 | CLK   | 0.09602   | 0.29404 | 4.57538 |
|                                 | CLK   | 0.08491   | 0.28312 | 4.56496 |

Internal switching power(pJ) to Q falling :

| Cell Name                       | Input | Power(pJ) |         |         |
|---------------------------------|-------|-----------|---------|---------|
|                                 |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__dffn_1 | CLK   | 0.14085   | 0.33034 | 4.67148 |
|                                 | CLK   | 0.12992   | 0.31911 | 4.64590 |

Internal switching power(pJ) to QN rising :

| Cell Name                       | Input | Power(pJ) |         |         |
|---------------------------------|-------|-----------|---------|---------|
|                                 |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__dffn_1 | CLK   | 0.14109   | 0.33043 | 4.66113 |
|                                 | CLK   | 0.13015   | 0.31849 | 4.63001 |

Internal switching power(pJ) to QN falling :

| Cell Name                       | Input | Power(pJ) |         |         |
|---------------------------------|-------|-----------|---------|---------|
|                                 |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__dffn_1 | CLK   | 0.09596   | 0.29574 | 4.55022 |
|                                 | CLK   | 0.08485   | 0.28472 | 4.55149 |

Passive power(pJ) for D rising (conditional):

| Cell Name                       | When                                | Power(pJ) |          |          |
|---------------------------------|-------------------------------------|-----------|----------|----------|
|                                 |                                     | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__dffn_1 | $(CLK * Q * !QN) + (CLK * !Q * QN)$ | 0.15510   | 0.88451  | 12.51980 |
|                                 | $(CLK * Q * !QN) + (CLK * !Q * QN)$ | 0.17659   | 0.90606  | 12.54120 |
|                                 | $!CLK$                              | -0.01323  | -0.01350 | -0.01342 |
|                                 | $!CLK$                              | 0.00666   | 0.00654  | 0.00646  |

Passive power(pJ) for D falling (conditional):

| Cell Name                       | When                                | Power(pJ) |          |          |
|---------------------------------|-------------------------------------|-----------|----------|----------|
|                                 |                                     | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__dffn_1 | $(CLK * Q * !QN) + (CLK * !Q * QN)$ | 0.23036   | 0.98617  | 12.90250 |
|                                 | $(CLK * Q * !QN) + (CLK * !Q * QN)$ | 0.20889   | 0.96464  | 12.88110 |
|                                 | $!CLK$                              | 0.01353   | 0.01364  | 0.01343  |
|                                 | $!CLK$                              | -0.00634  | -0.00648 | -0.00645 |

Passive power(pJ) for CLK rising (conditional):

| Cell Name                       | When             | Power(pJ) |         |          |
|---------------------------------|------------------|-----------|---------|----------|
|                                 |                  | first     | mid     | last     |
| gf180mcu_osu_sc_gp9t3v3__dffn_1 | $(D * Q * !QN)$  | 0.03606   | 0.43294 | 7.27886  |
|                                 | $(D * Q * !QN)$  | 0.05822   | 0.45522 | 7.30110  |
|                                 | $(D * !Q * QN)$  | 0.20003   | 0.56828 | 19.85620 |
|                                 | $(D * !Q * QN)$  | 0.22298   | 0.59129 | 19.87950 |
|                                 | $(!D * Q * !QN)$ | 0.23245   | 0.53785 | 7.53163  |
|                                 | $(!D * Q * !QN)$ | 0.25380   | 0.55923 | 7.55303  |
|                                 | $(!D * !Q * QN)$ | 0.04547   | 0.44357 | 7.28931  |
|                                 | $(!D * !Q * QN)$ | 0.06724   | 0.46546 | 7.31111  |

Passive power(pJ) for CLK falling (conditional):

| Cell Name                       | When           | Power(pJ) |         |         |
|---------------------------------|----------------|-----------|---------|---------|
|                                 |                | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__dffn_1 | (D * Q * !QN)  | 0.06209   | 0.46132 | 7.29869 |
|                                 | (D * Q * !QN)  | 0.04004   | 0.43912 | 7.27624 |
|                                 | (!D * !Q * QN) | 0.06732   | 0.46706 | 7.30729 |
|                                 | (!D * !Q * QN) | 0.04546   | 0.44519 | 7.28546 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_DFFSR\_1

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT |   |   |     | OUTPUT |     |
|-------|---|---|-----|--------|-----|
| D     | R | S | CLK | Q      | QN  |
| 0     | 1 | 1 | R   | 0      | 1   |
| 1     | 1 | 1 | R   | 1      | 0   |
| x     | 0 | x | x   | 0      | 1   |
| x     | 1 | 0 | x   | 1      | 0   |
| x     | 1 | 1 | x   | IQ     | IQN |

## Footprint

| Cell Name                        | Area      |
|----------------------------------|-----------|
| gf180mcu_osu_sc_gp9t3v3__dffsr_1 | 130.41000 |

## Pin Capacitance Information

| Cell Name                        | Pin Cap(pf) |         |         |         | Max Cap(pf) |          |
|----------------------------------|-------------|---------|---------|---------|-------------|----------|
|                                  | D           | R       | S       | CLK     | Q           | QN       |
| gf180mcu_osu_sc_gp9t3v3__dffsr_1 | 0.00356     | 0.00406 | 0.00810 | 0.01419 | 15.67782    | 15.73825 |

## Leakage Information

| Cell Name                        | Leakage(nW) |         |         |
|----------------------------------|-------------|---------|---------|
|                                  | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__dffsr_1 | 0.00000     | 0.00694 | 0.00849 |

## Delay Information

Delay(ns) to Q rising :

| Cell Name                        | Timing Arc(Dir) | Delay(ns) |         |           |
|----------------------------------|-----------------|-----------|---------|-----------|
|                                  |                 | First     | Mid     | Last      |
| gf180mcu_osu_sc_gp9t3v3__dffsr_1 | CLK->Q (RR)     | 0.31830   | 4.12101 | 157.35500 |
|                                  | QN->Q (FR)      | 0.03869   | 3.45702 | 102.66000 |
|                                  | R->Q (RR)       | 0.25355   | 4.16180 | 160.47000 |
|                                  | S->Q (FR)       | 0.22901   | 4.63066 | 168.17400 |

Delay(ns) to Q falling :

| Cell Name                        | Timing Arc(Dir) | Delay(ns) |         |           |
|----------------------------------|-----------------|-----------|---------|-----------|
|                                  |                 | First     | Mid     | Last      |
| gf180mcu_osu_sc_gp9t3v3__dffsr_1 | CLK->Q (RF)     | 0.36468   | 4.12794 | 156.65000 |
|                                  | QN->Q (RF)      | 0.03096   | 2.69307 | 87.48170  |
|                                  | R->Q (FF)       | 0.21405   | 4.74494 | 169.98000 |

Delay(ns) to QN rising :

| Cell Name                        | Timing Arc(Dir) | Delay(ns) |         |          |
|----------------------------------|-----------------|-----------|---------|----------|
|                                  |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__dffsr_1 | CLK->QN (RR)    | 0.33455   | 1.90830 | 63.85290 |
|                                  | R->QN (FR)      | 0.18413   | 2.52620 | 77.17290 |

Delay(ns) to QN falling :

| Cell Name                        | Timing Arc(Dir) | Delay(ns) |         |          |
|----------------------------------|-----------------|-----------|---------|----------|
|                                  |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__dffsr_1 | CLK->QN (RF)    | 0.28250   | 1.65931 | 54.34610 |
|                                  | R->QN (RF)      | 0.21815   | 1.69939 | 57.44750 |
|                                  | S->QN (FF)      | 0.19371   | 2.16858 | 65.11800 |

## Constraint Information

Constraints(ns) for D rising :

| Cell Name                        | Timing Check | Ref Pin(trans) | Reference Slew Rate(ns) |          |          |
|----------------------------------|--------------|----------------|-------------------------|----------|----------|
|                                  |              |                | first                   | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__dffsr_1 | hold         | CLK (R)        | -0.15300                | -0.21883 | 2.11172  |
|                                  | setup        | CLK (R)        | 0.26990                 | 0.63601  | -1.94971 |

Constraints(ns) for D falling :

| Cell Name                        | Timing Check | Ref Pin(trans) | Reference Slew Rate(ns) |          |           |
|----------------------------------|--------------|----------------|-------------------------|----------|-----------|
|                                  |              |                | first                   | mid      | last      |
| gf180mcu_osu_sc_gp9t3v3__dffsr_1 | hold         | CLK (R)        | -0.17231                | -1.28868 | -44.38480 |
|                                  | setup        | CLK (R)        | 0.20049                 | 1.31700  | 44.38900  |

Constraints(ns) for D rising (conditional):

| Cell Name                        | Timing Check | Ref Pin(trans) | When    | Reference Slew Rate(ns) |          |          |
|----------------------------------|--------------|----------------|---------|-------------------------|----------|----------|
|                                  |              |                |         | first                   | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__dffsr_1 | hold         | CLK (R)        | (R * S) | -0.15300                | -0.21883 | 2.11172  |
|                                  | setup        | CLK (R)        | (R * S) | 0.26990                 | 0.63601  | -1.94971 |

Constraints(ns) for D falling (conditional):

| Cell Name                        | Timing Check | Ref Pin(trans) | When    | Reference Slew Rate(ns) |          |           |
|----------------------------------|--------------|----------------|---------|-------------------------|----------|-----------|
|                                  |              |                |         | first                   | mid      | last      |
| gf180mcu_osu_sc_gp9t3v3__dffsr_1 | hold         | CLK (R)        | (R * S) | -0.17231                | -1.28868 | -44.38480 |
|                                  | setup        | CLK (R)        | (R * S) | 0.20049                 | 1.31700  | 44.38900  |

Constraints(ns) for R rising :



| Cell Name                        | Timing Check | Ref Pin(trans) | Reference Slew Rate(ns) |          |          |
|----------------------------------|--------------|----------------|-------------------------|----------|----------|
|                                  |              |                | first                   | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__dffsr_1 | recovery     | CLK (R)        | 0.15874                 | 0.52977  | 4.02928  |
|                                  | removal      | CLK (R)        | -0.02921                | -0.05471 | -0.58999 |
|                                  | hold         | S (R)          | -0.18548                | -0.56532 | 0.08184  |
|                                  | setup        | S (R)          | 0.23292                 | 1.21814  | 26.70080 |

Constraints(ns) for R rising (conditional):

| Cell Name                        | Timing Check | Ref Pin(trans) | When    | Reference Slew Rate(ns) |          |          |
|----------------------------------|--------------|----------------|---------|-------------------------|----------|----------|
|                                  |              |                |         | first                   | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__dffsr_1 | recovery     | CLK (R)        | (D * S) | 0.15874                 | 0.52977  | 4.02928  |
|                                  | removal      | CLK (R)        | (D * S) | -0.02921                | -0.05471 | -0.58999 |
|                                  | hold         | S (R)          | CLK     | -0.18615                | -0.56532 | 0.08184  |
|                                  | hold         | S (R)          | !CLK    | -0.18548                | -0.57748 | -0.22940 |
|                                  | setup        | S (R)          | CLK     | 0.21042                 | 1.19719  | 22.74950 |
|                                  | setup        | S (R)          | !CLK    | 0.23292                 | 1.21814  | 26.70080 |

Constraints(ns) for R falling (conditional):

| Cell Name                        | Timing Check    | Ref Pin(trans) | When       | Reference Slew Rate(ns) |         |           |
|----------------------------------|-----------------|----------------|------------|-------------------------|---------|-----------|
|                                  |                 |                |            | first                   | mid     | last      |
| gf180mcu_osu_sc_gp9t3v3__dffsr_1 | min_pulse_width | R ()           | (CLK * S)  | 0.14220                 | 4.36523 | 165.00100 |
|                                  | min_pulse_width | R ()           | (!CLK * S) | 0.13743                 | 4.36523 | 165.00100 |

Constraints(ns) for S rising :

| Cell Name                        | Timing Check | Ref Pin(trans) | Reference Slew Rate(ns) |          |          |
|----------------------------------|--------------|----------------|-------------------------|----------|----------|
|                                  |              |                | first                   | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__dffsr_1 | recovery     | CLK (R)        | 0.06844                 | 0.95317  | 11.54640 |
|                                  | removal      | CLK (R)        | -0.03967                | -0.21579 | -5.83566 |

**Constraints(ns) for S rising (conditional):**

| Cell Name                        | Timing Check | Ref Pin(trans) | When     | Reference Slew Rate(ns) |          |          |
|----------------------------------|--------------|----------------|----------|-------------------------|----------|----------|
|                                  |              |                |          | first                   | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__dffsr_1 | recovery     | CLK (R)        | (!D * R) | 0.06844                 | 0.95317  | 11.54640 |
|                                  | removal      | CLK (R)        | (!D * R) | -0.03967                | -0.21579 | -5.83566 |

**Constraints(ns) for S falling (conditional):**

| Cell Name                        | Timing Check    | Ref Pin(trans) | When       | Reference Slew Rate(ns) |         |           |
|----------------------------------|-----------------|----------------|------------|-------------------------|---------|-----------|
|                                  |                 |                |            | first                   | mid     | last      |
| gf180mcu_osu_sc_gp9t3v3__dffsr_1 | min_pulse_width | S ()           | (CLK * R)  | 0.18751                 | 4.36523 | 165.00100 |
|                                  | min_pulse_width | S ()           | (!CLK * R) | 0.20897                 | 4.36523 | 165.00100 |

**Constraints(ns) for CLK rising (conditional):**

| Cell Name                        | Timing Check    | Ref Pin(trans) | When         | Reference Slew Rate(ns) |         |           |
|----------------------------------|-----------------|----------------|--------------|-------------------------|---------|-----------|
|                                  |                 |                |              | first                   | mid     | last      |
| gf180mcu_osu_sc_gp9t3v3__dffsr_1 | min_pulse_width | CLK ()         | (D * R * S)  | 0.16128                 | 4.36523 | 165.00100 |
|                                  | min_pulse_width | CLK ()         | (!D * R * S) | 0.16843                 | 4.36523 | 165.00100 |

**Constraints(ns) for CLK falling (conditional):**

| Cell Name                        | Timing Check    | Ref Pin(trans) | When         | Reference Slew Rate(ns) |         |           |
|----------------------------------|-----------------|----------------|--------------|-------------------------|---------|-----------|
|                                  |                 |                |              | first                   | mid     | last      |
| gf180mcu_osu_sc_gp9t3v3__dffsr_1 | min_pulse_width | CLK ()         | (D * R * S)  | 0.31867                 | 4.36523 | 165.00100 |
|                                  | min_pulse_width | CLK ()         | (!D * R * S) | 0.17559                 | 4.36523 | 165.00100 |

## Power Information

Internal switching power(pJ) to Q rising :

| Cell Name                        | Input | Power(pJ) |         |         |
|----------------------------------|-------|-----------|---------|---------|
|                                  |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__dffsr_1 | CLK   | 0.04676   | 0.34344 | 5.55587 |
|                                  | CLK   | 0.07169   | 0.36837 | 5.59273 |
|                                  | R     | 0.06747   | 0.29077 | 4.70304 |
|                                  | R     | 0.09917   | 0.32168 | 4.75503 |
|                                  | S     | 0.07959   | 0.34692 | 5.10845 |
|                                  | S     | 0.06339   | 0.33070 | 5.10156 |

Internal switching power(pJ) to Q falling :

| Cell Name                        | Input | Power(pJ) |         |         |
|----------------------------------|-------|-----------|---------|---------|
|                                  |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__dffsr_1 | CLK   | 0.05056   | 0.25953 | 4.52387 |
|                                  | CLK   | 0.07507   | 0.28410 | 4.54822 |
|                                  | R     | 0.10018   | 0.33241 | 4.85169 |
|                                  | R     | 0.08335   | 0.31512 | 4.82412 |

Internal switching power(pJ) to QN rising :

| Cell Name                        | Input | Power(pJ) |         |         |
|----------------------------------|-------|-----------|---------|---------|
|                                  |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__dffsr_1 | CLK   | 0.05053   | 0.25911 | 4.50719 |
|                                  | CLK   | 0.07503   | 0.28368 | 4.53460 |
|                                  | R     | 0.10016   | 0.33254 | 4.83769 |
|                                  | R     | 0.08333   | 0.31512 | 4.80068 |

Internal switching power(pJ) to QN falling :

| Cell Name                        | Input | Power(pJ) |         |         |
|----------------------------------|-------|-----------|---------|---------|
|                                  |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__dffsr_1 | CLK   | 0.04667   | 0.34316 | 5.52212 |
|                                  | CLK   | 0.07160   | 0.36806 | 5.55461 |
|                                  | R     | 0.08230   | 0.30455 | 4.69229 |
|                                  | R     | 0.09907   | 0.32127 | 4.70903 |
|                                  | S     | 0.07951   | 0.34690 | 5.07347 |
|                                  | S     | 0.06331   | 0.33075 | 5.08153 |

Passive power(pJ) for D rising (conditional):

| Cell Name                        | When                                                   | Power(pJ) |          |          |
|----------------------------------|--------------------------------------------------------|-----------|----------|----------|
|                                  |                                                        | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__dffsr_1 | CLK                                                    | -0.01321  | -0.01345 | -0.01329 |
|                                  | CLK                                                    | 0.00462   | 0.00466  | 0.00461  |
|                                  | (!CLK * R * S * Q * !QN)<br>+ (!CLK * R * S * !Q * QN) | 0.06359   | 0.35477  | 5.69610  |
|                                  | (!CLK * R * S * Q * !QN)<br>+ (!CLK * R * S * !Q * QN) | 0.08717   | 0.37835  | 5.71946  |
|                                  | (!CLK * R * !S * Q * !QN)                              | 0.02346   | 0.29114  | 5.26996  |
|                                  | (!CLK * R * !S * Q * !QN)                              | 0.05321   | 0.32095  | 5.29961  |
|                                  | (!CLK * !R * S * !Q * QN)                              | 0.02319   | 0.29026  | 5.27033  |
|                                  | (!CLK * !R * S * !Q * QN)                              | 0.05300   | 0.32023  | 5.30018  |
|                                  | (!CLK * !R * !S * !Q * QN)                             | 0.02343   | 0.29114  | 5.26997  |
|                                  | (!CLK * !R * !S * !Q * QN)                             | 0.05318   | 0.32095  | 5.29961  |

Passive power(pJ) for D falling (conditional):

| Cell Name                        | When                                                   | Power(pJ) |          |          |
|----------------------------------|--------------------------------------------------------|-----------|----------|----------|
|                                  |                                                        | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__dffsr_1 | CLK                                                    | 0.01339   | 0.01350  | 0.01329  |
|                                  | CLK                                                    | -0.00442  | -0.00457 | -0.00455 |
|                                  | (!CLK * R * S * Q * !QN)<br>+ (!CLK * R * S * !Q * QN) | 0.08497   | 0.38335  | 5.75601  |
|                                  | (!CLK * R * S * Q * !QN)<br>+ (!CLK * R * S * !Q * QN) | 0.06150   | 0.35987  | 5.73264  |
|                                  | (!CLK * R * !S * Q * !QN)                              | 0.03875   | 0.30815  | 5.28734  |
|                                  | (!CLK * R * !S * Q * !QN)                              | 0.00910   | 0.27845  | 5.25786  |
|                                  | (!CLK * !R * S * !Q * QN)                              | 0.03904   | 0.30815  | 5.28778  |
|                                  | (!CLK * !R * S * !Q * QN)                              | 0.00930   | 0.27844  | 5.25817  |
|                                  | (!CLK * !R * !S * !Q * QN)                             | 0.03874   | 0.30814  | 5.28786  |
|                                  | (!CLK * !R * !S * !Q * QN)                             | 0.00909   | 0.27844  | 5.25809  |

Passive power(pJ) for R rising (conditional):

| Cell Name                        | When                                               | Power(pJ) |         |         |
|----------------------------------|----------------------------------------------------|-----------|---------|---------|
|                                  |                                                    | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__dffsr_1 | (CLK * S * !Q * QN) +<br>(!CLK * !D * S * !Q * QN) | 0.00539   | 0.37280 | 6.86143 |
|                                  | (CLK * S * !Q * QN) +<br>(!CLK * !D * S * !Q * QN) | 0.02765   | 0.39515 | 6.88370 |
|                                  | (!CLK * D * S * !Q * QN)                           | 0.04422   | 0.42877 | 7.19873 |
|                                  | (!CLK * D * S * !Q * QN)                           | 0.06107   | 0.44569 | 7.21577 |

Passive power(pJ) for R falling (conditional):

| Cell Name                        | When                                              | Power(pJ) |         |         |
|----------------------------------|---------------------------------------------------|-----------|---------|---------|
|                                  |                                                   | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__dffsr_1 | $(CLK * S * !Q * QN) + (!CLK * !D * S * !Q * QN)$ | 0.03545   | 0.40774 | 6.89545 |
|                                  | $(CLK * S * !Q * QN) + (!CLK * !D * S * !Q * QN)$ | 0.01332   | 0.38542 | 6.87329 |
|                                  | $(!CLK * D * S * !Q * QN)$                        | 0.07284   | 0.46456 | 7.22421 |
|                                  | $(!CLK * D * S * !Q * QN)$                        | 0.05601   | 0.44778 | 7.20680 |

Passive power(pJ) for S rising (conditional):

| Cell Name                        | When                                             | Power(pJ) |          |          |
|----------------------------------|--------------------------------------------------|-----------|----------|----------|
|                                  |                                                  | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__dffsr_1 | $(CLK * R * Q * !QN) + (!CLK * D * R * Q * !QN)$ | -0.02786  | -0.02797 | -0.02819 |
|                                  | $(CLK * R * Q * !QN) + (!CLK * D * R * Q * !QN)$ | 0.00382   | 0.00385  | 0.00366  |
|                                  | $(!R * !Q * QN)$                                 | -0.02611  | -0.02708 | -0.02697 |
|                                  | $(!R * !Q * QN)$                                 | 0.01374   | 0.01310  | 0.01305  |
|                                  | $(!CLK * !D * R * Q * !QN)$                      | 0.01824   | 0.30474  | 5.57959  |
|                                  | $(!CLK * !D * R * Q * !QN)$                      | 0.05575   | 0.34239  | 5.61713  |

Passive power(pJ) for S falling (conditional):

| Cell Name                        | When                                             | Power(pJ) |          |          |
|----------------------------------|--------------------------------------------------|-----------|----------|----------|
|                                  |                                                  | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__dffsr_1 | $(CLK * R * Q * !QN) + (!CLK * D * R * Q * !QN)$ | 0.02858   | 0.02853  | 0.02841  |
|                                  | $(CLK * R * Q * !QN) + (!CLK * D * R * Q * !QN)$ | -0.00354  | -0.00352 | -0.00349 |
|                                  | $(!R * !Q * QN)$                                 | 0.02716   | 0.02725  | 0.02697  |
|                                  | $(!R * !Q * QN)$                                 | -0.01297  | -0.01295 | -0.01296 |
|                                  | $(!CLK * !D * R * Q * !QN)$                      | 0.05862   | 0.34157  | 5.61922  |
|                                  | $(!CLK * !D * R * Q * !QN)$                      | 0.02098   | 0.30368  | 5.58160  |

Passive power(pJ) for CLK rising (conditional):

| Cell Name                        | When                                           | Power(pJ) |         |          |
|----------------------------------|------------------------------------------------|-----------|---------|----------|
|                                  |                                                | first     | mid     | last     |
| gf180mcu_osu_sc_gp9t3v3__dffsr_1 | $(D * R * Q * !QN)$                            | -0.00398  | 0.36522 | 6.85435  |
|                                  | $(D * R * Q * !QN)$                            | 0.04258   | 0.41189 | 6.90085  |
|                                  | $(D * !R * S * !Q * QN)$                       | 0.02413   | 0.40119 | 7.11611  |
|                                  | $(D * !R * S * !Q * QN)$                       | 0.06787   | 0.44390 | 7.15709  |
|                                  | $(D * !R * !S * !Q * QN)$                      | 0.02385   | 0.40130 | 7.11686  |
|                                  | $(D * !R * !S * !Q * QN)$                      | 0.06767   | 0.44374 | 7.15667  |
|                                  | $(!D * R * S * !Q * QN) + (!D * !R * !Q * QN)$ | -0.00508  | 0.35989 | 6.85076  |
|                                  | $(!D * R * S * !Q * QN) + (!D * !R * !Q * QN)$ | 0.04869   | 0.41391 | 6.90453  |
|                                  | $(!D * R * !S * Q * !QN)$                      | 0.01379   | 0.63391 | 11.70040 |
|                                  | $(!D * R * !S * Q * !QN)$                      | 0.06996   | 0.69059 | 11.75680 |

Passive power(pJ) for CLK falling (conditional):

| Cell Name                        | When                                                | Power(pJ) |         |          |
|----------------------------------|-----------------------------------------------------|-----------|---------|----------|
|                                  |                                                     | first     | mid     | last     |
| gf180mcu_osu_sc_gp9t3v3__dffsr_1 | $(D * R * S * !Q * QN)$                             | 0.12430   | 0.56818 | 9.66702  |
|                                  | $(D * R * S * !Q * QN)$                             | 0.07699   | 0.51958 | 9.59693  |
|                                  | $(D * R * Q * !QN)$                                 | 0.04095   | 0.41164 | 6.90247  |
|                                  | $(D * R * Q * !QN)$                                 | -0.00532  | 0.36516 | 6.85593  |
|                                  | $(D * !R * S * !Q * QN)$                            | 0.08037   | 0.45763 | 7.15996  |
|                                  | $(D * !R * S * !Q * QN)$                            | 0.03664   | 0.41381 | 7.11500  |
|                                  | $(D * !R * !S * !Q * QN)$                           | 0.08079   | 0.45756 | 7.15722  |
|                                  | $(D * !R * !S * !Q * QN)$                           | 0.03688   | 0.41421 | 7.11333  |
|                                  | $(!D * R * S * Q * !QN)$                            | 0.10990   | 0.69691 | 9.83628  |
|                                  | $(!D * R * S * Q * !QN)$                            | 0.05955   | 0.64625 | 9.78469  |
|                                  | $(!D * R * S * !Q * QN) +$<br>$(!D * !R * !Q * QN)$ | 0.05021   | 0.42143 | 6.90589  |
|                                  | $(!D * R * S * !Q * QN) +$<br>$(!D * !R * !Q * QN)$ | -0.00393  | 0.36719 | 6.85195  |
|                                  | $(!D * R * !S * Q * !QN)$                           | 0.05931   | 0.69122 | 11.75010 |
|                                  | $(!D * R * !S * Q * !QN)$                           | 0.00299   | 0.63468 | 11.69380 |



# GF180MCU\_OSU\_SC\_GP9T3V3\_\_DFF\_1

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT |     | OUTPUT |     |
|-------|-----|--------|-----|
| D     | CLK | Q      | QN  |
| 0     | R   | 0      | 1   |
| 1     | R   | 1      | 0   |
| x     | x   | IQ     | IQN |

## Footprint

| Cell Name                      | Area     |
|--------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__dff_1 | 92.61000 |

## Pin Capacitance Information

| Cell Name                      | Pin Cap(pf) |         | Max Cap(pf) |          |
|--------------------------------|-------------|---------|-------------|----------|
|                                | D           | CLK     | Q           | QN       |
| gf180mcu_osu_sc_gp9t3v3__dff_1 | 0.00394     | 0.01204 | 15.73103    | 15.73823 |

## Leakage Information

| Cell Name                      | Leakage(nW) |              |              |
|--------------------------------|-------------|--------------|--------------|
|                                | Min.        | Avg          | Max.         |
| gf180mcu_osu_sc_gp9t3v3__dff_1 | 0.00000     | 100780.00000 | 174312.00000 |

## Delay Information

Delay(ns) to Q rising :

| Cell Name                      | Timing Arc(Dir) | Delay(ns) |         |           |
|--------------------------------|-----------------|-----------|---------|-----------|
|                                |                 | First     | Mid     | Last      |
| gf180mcu_osu_sc_gp9t3v3__dff_1 | CLK->Q (RR)     | 0.27143   | 2.34546 | 118.43500 |
|                                | QN->Q (FR)      | 0.03869   | 3.46088 | 102.88800 |

Delay(ns) to Q falling :

| Cell Name                      | Timing Arc(Dir) | Delay(ns) |         |           |
|--------------------------------|-----------------|-----------|---------|-----------|
|                                |                 | First     | Mid     | Last      |
| gf180mcu_osu_sc_gp9t3v3__dff_1 | CLK->Q (RF)     | 0.55873   | 4.07015 | 156.99700 |
|                                | QN->Q (RF)      | 0.03096   | 2.69716 | 87.68530  |

Delay(ns) to QN rising :

| Cell Name                      | Timing Arc(Dir) | Delay(ns) |         |          |
|--------------------------------|-----------------|-----------|---------|----------|
|                                |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__dff_1 | CLK->QN (RR)    | 0.52999   | 1.84289 | 63.63520 |

Delay(ns) to QN falling :

| Cell Name                      | Timing Arc(Dir) | Delay(ns) |          |          |
|--------------------------------|-----------------|-----------|----------|----------|
|                                |                 | First     | Mid      | Last     |
| gf180mcu_osu_sc_gp9t3v3__dff_1 | CLK->QN (RF)    | 0.23822   | -0.12283 | 14.78590 |

## Constraint Information

Constraints(ns) for D rising :

| Cell Name                      | Timing Check | Ref Pin(trans) | Reference Slew Rate(ns) |          |           |
|--------------------------------|--------------|----------------|-------------------------|----------|-----------|
|                                |              |                | first                   | mid      | last      |
| gf180mcu_osu_sc_gp9t3v3__dff_1 | hold         | CLK (R)        | -0.07478                | -0.50537 | -11.39580 |
|                                | setup        | CLK (R)        | 0.38037                 | 1.97359  | 61.18600  |

Constraints(ns) for D falling :

| Cell Name                      | Timing Check | Ref Pin(trans) | Reference Slew Rate(ns) |         |           |
|--------------------------------|--------------|----------------|-------------------------|---------|-----------|
|                                |              |                | first                   | mid     | last      |
| gf180mcu_osu_sc_gp9t3v3__dff_1 | hold         | CLK (R)        | -0.38172                | 0.97867 | 81.80590  |
|                                | setup        | CLK (R)        | 0.51469                 | 3.46512 | 122.23100 |

Constraints(ns) for CLK rising (conditional):

| Cell Name                      | Timing Check    | Ref Pin(trans) | When | Reference Slew Rate(ns) |         |           |
|--------------------------------|-----------------|----------------|------|-------------------------|---------|-----------|
|                                |                 |                |      | first                   | mid     | last      |
| gf180mcu_osu_sc_gp9t3v3__dff_1 | min_pulse_width | CLK ()         | D    | 0.19943                 | 4.36523 | 165.00100 |
|                                | min_pulse_width | CLK ()         | !D   | 0.11597                 | 4.36523 | 165.00100 |

Constraints(ns) for CLK falling (conditional):

| Cell Name                      | Timing Check    | Ref Pin(trans) | When | Reference Slew Rate(ns) |         |           |
|--------------------------------|-----------------|----------------|------|-------------------------|---------|-----------|
|                                |                 |                |      | first                   | mid     | last      |
| gf180mcu_osu_sc_gp9t3v3__dff_1 | min_pulse_width | CLK ()         | D    | 0.39737                 | 4.36523 | 288.85400 |
|                                | min_pulse_width | CLK ()         | !D   | 0.35683                 | 4.50905 | 284.04100 |

## Power Information

Internal switching power(pJ) to Q rising :

| Cell Name                      | Input | Power(pJ) |         |         |
|--------------------------------|-------|-----------|---------|---------|
|                                |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__dff_1 | CLK   | 0.06172   | 0.25540 | 4.47452 |
|                                | CLK   | 0.08468   | 0.27811 | 4.49723 |

Internal switching power(pJ) to Q falling :

| Cell Name                      | Input | Power(pJ) |         |         |
|--------------------------------|-------|-----------|---------|---------|
|                                |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__dff_1 | CLK   | 0.10669   | 0.26588 | 4.52267 |
|                                | CLK   | 0.13219   | 0.29181 | 4.54822 |

Internal switching power(pJ) to QN rising :

| Cell Name                      | Input | Power(pJ) |         |         |
|--------------------------------|-------|-----------|---------|---------|
|                                |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__dff_1 | CLK   | 0.10670   | 0.26594 | 4.51837 |
|                                | CLK   | 0.13220   | 0.29160 | 4.53455 |

Internal switching power(pJ) to QN falling :

| Cell Name                      | Input | Power(pJ) |         |         |
|--------------------------------|-------|-----------|---------|---------|
|                                |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__dff_1 | CLK   | 0.06162   | 0.25614 | 4.46849 |
|                                | CLK   | 0.08458   | 0.27789 | 4.49535 |

Passive power(pJ) for D rising (conditional):

| Cell Name                      | When                                                                              | Power(pJ) |          |          |
|--------------------------------|-----------------------------------------------------------------------------------|-----------|----------|----------|
|                                |                                                                                   | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__dff_1 | CLK                                                                               | -0.01304  | -0.01345 | -0.01329 |
|                                | CLK                                                                               | 0.00666   | 0.00654  | 0.00646  |
|                                | $(\neg \text{CLK} * Q * \neg \text{QN}) + (\neg \text{CLK} * \neg Q * \text{QN})$ | 0.15146   | 0.88072  | 12.51460 |
|                                | $(\neg \text{CLK} * Q * \neg \text{QN}) + (\neg \text{CLK} * \neg Q * \text{QN})$ | 0.19225   | 0.92147  | 12.55560 |

Passive power(pJ) for D falling (conditional):

| Cell Name                      | When                                                                              | Power(pJ) |          |          |
|--------------------------------|-----------------------------------------------------------------------------------|-----------|----------|----------|
|                                |                                                                                   | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__dff_1 | CLK                                                                               | 0.01337   | 0.01349  | 0.01329  |
|                                | CLK                                                                               | -0.00634  | -0.00648 | -0.00645 |
|                                | $(\neg \text{CLK} * Q * \neg \text{QN}) + (\neg \text{CLK} * \neg Q * \text{QN})$ | 0.22903   | 0.98590  | 12.88440 |
|                                | $(\neg \text{CLK} * Q * \neg \text{QN}) + (\neg \text{CLK} * \neg Q * \text{QN})$ | 0.18849   | 0.94520  | 12.84680 |

Passive power(pJ) for CLK rising (conditional):

| Cell Name                      | When                            | Power(pJ) |         |         |
|--------------------------------|---------------------------------|-----------|---------|---------|
|                                |                                 | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__dff_1 | $(D * Q * \neg \text{QN})$      | -0.00228  | 0.35978 | 6.85299 |
|                                | $(D * Q * \neg \text{QN})$      | 0.04256   | 0.40434 | 6.89739 |
|                                | $(\neg D * \neg Q * \text{QN})$ | -0.00487  | 0.35952 | 6.85053 |
|                                | $(\neg D * \neg Q * \text{QN})$ | 0.04786   | 0.41276 | 6.90365 |

Passive power(pJ) for CLK falling (conditional):

| Cell Name                      | When           | Power(pJ) |         |         |
|--------------------------------|----------------|-----------|---------|---------|
|                                |                | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__dff_1 | (D * Q * !QN)  | 0.04188   | 0.41267 | 6.90383 |
|                                | (D * Q * !QN)  | -0.00268  | 0.36808 | 6.85911 |
|                                | (D * !Q * QN)  | 0.20322   | 0.60369 | 7.35470 |
|                                | (D * !Q * QN)  | 0.17120   | 0.57459 | 7.31782 |
|                                | (!D * Q * !QN) | 0.22886   | 0.65776 | 7.53008 |
|                                | (!D * Q * !QN) | 0.16491   | 0.59275 | 7.46490 |
|                                | (!D * !Q * QN) | 0.05086   | 0.41266 | 6.90460 |
|                                | (!D * !Q * QN) | -0.00211  | 0.35933 | 6.85169 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_DLATN\_1

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT |     | OUTPUT |
|-------|-----|--------|
| D     | CLK | Q      |
| 0     | 0   | 0      |
| x     | 1   | IQ     |
| 1     | 0   | 1      |

## Footprint

| Cell Name                        | Area     |
|----------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__dlatn_1 | 71.82000 |

## Pin Capacitance Information

| Cell Name                        | Pin Cap(pf) |         | Max Cap(pf) |
|----------------------------------|-------------|---------|-------------|
|                                  | D           | CLK     | Q           |
| gf180mcu_osu_sc_gp9t3v3__dlatn_1 | 0.00395     | 0.00405 | 15.49329    |

## Leakage Information

| Cell Name                        | Leakage(nW) |         |         |
|----------------------------------|-------------|---------|---------|
|                                  | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__dlatn_1 | 0.00000     | 0.00481 | 0.00529 |

## Delay Information

Delay(ns) to Q rising :

| Cell Name                        | Timing Arc(Dir) | Delay(ns) |         |          |
|----------------------------------|-----------------|-----------|---------|----------|
|                                  |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__dlatn_1 | CLK->Q (FR)     | 0.28814   | 2.62214 | 76.23070 |
|                                  | D->Q (RR)       | 0.24486   | 1.79776 | 62.41010 |

Delay(ns) to Q falling :

| Cell Name                        | Timing Arc(Dir) | Delay(ns) |         |          |
|----------------------------------|-----------------|-----------|---------|----------|
|                                  |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__dlatn_1 | CLK->Q (FF)     | 0.33959   | 2.44254 | 68.97380 |
|                                  | D->Q (FF)       | 0.27113   | 2.42533 | 69.80940 |



## Constraint Information

Constraints(ns) for D rising :

| Cell Name                        | Timing Check | Ref Pin(trans) | Reference Slew Rate(ns) |          |          |
|----------------------------------|--------------|----------------|-------------------------|----------|----------|
|                                  |              |                | first                   | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__dlatn_1 | hold         | CLK (R)        | -0.08483                | -0.24923 | -4.00732 |
|                                  | setup        | CLK (R)        | 0.09270                 | 0.28778  | 6.99863  |

Constraints(ns) for D falling :

| Cell Name                        | Timing Check | Ref Pin(trans) | Reference Slew Rate(ns) |          |          |
|----------------------------------|--------------|----------------|-------------------------|----------|----------|
|                                  |              |                | first                   | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__dlatn_1 | hold         | CLK (R)        | -0.07587                | -0.38296 | -6.15850 |
|                                  | setup        | CLK (R)        | 0.08715                 | 0.38600  | 6.16953  |

Constraints(ns) for CLK falling (conditional):

| Cell Name                        | Timing Check    | Ref Pin(trans) | When | Reference Slew Rate(ns) |         |           |
|----------------------------------|-----------------|----------------|------|-------------------------|---------|-----------|
|                                  |                 |                |      | first                   | mid     | last      |
| gf180mcu_osu_sc_gp9t3v3__dlatn_1 | min_pulse_width | CLK ()         | D    | 0.14458                 | 4.36523 | 165.00100 |
|                                  | min_pulse_width | CLK ()         | !D   | 0.15889                 | 4.36523 | 165.00100 |

## Power Information

Internal switching power(pJ) to Q rising :

| Cell Name                        | Input | Power(pJ) |         |         |
|----------------------------------|-------|-----------|---------|---------|
|                                  |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__dlatn_1 | CLK   | 0.13037   | 0.55935 | 7.68051 |
|                                  | CLK   | 0.10930   | 0.53763 | 7.65200 |
|                                  | D     | 0.07488   | 0.43451 | 6.81991 |
|                                  | D     | 0.09637   | 0.45565 | 6.80578 |

Internal switching power(pJ) to Q falling :

| Cell Name                        | Input | Power(pJ) |         |         |
|----------------------------------|-------|-----------|---------|---------|
|                                  |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__dlatn_1 | CLK   | 0.13619   | 0.53592 | 7.36956 |
|                                  | CLK   | 0.11376   | 0.51398 | 7.35564 |
|                                  | D     | 0.10349   | 0.46402 | 6.79828 |
|                                  | D     | 0.08194   | 0.44301 | 6.77862 |

Passive power(pJ) for D rising (conditional):

| Cell Name                        | When | Power(pJ) |          |          |
|----------------------------------|------|-----------|----------|----------|
|                                  |      | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__dlatn_1 | CLK  | -0.01279  | -0.01360 | -0.01343 |
|                                  | CLK  | 0.00707   | 0.00663  | 0.00649  |

Passive power(pJ) for D falling (conditional):

| Cell Name                        | When | Power(pJ) |          |          |
|----------------------------------|------|-----------|----------|----------|
|                                  |      | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__dlatn_1 | CLK  | 0.01345   | 0.01366  | 0.01343  |
|                                  | CLK  | -0.00640  | -0.00648 | -0.00644 |

Passive power(pJ) for CLK rising (conditional):

| Cell Name                        | When      | Power(pJ) |         |         |
|----------------------------------|-----------|-----------|---------|---------|
|                                  |           | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__dlatn_1 | (D * Q)   | 0.02705   | 0.42671 | 7.27341 |
|                                  | (D * Q)   | 0.04879   | 0.44844 | 7.29517 |
|                                  | (!D * !Q) | 0.03005   | 0.43039 | 7.27672 |
|                                  | (!D * !Q) | 0.05212   | 0.45262 | 7.29872 |

Passive power(pJ) for CLK falling (conditional):

| Cell Name                        | When      | Power(pJ) |         |         |
|----------------------------------|-----------|-----------|---------|---------|
|                                  |           | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__dlatn_1 | (D * Q)   | 0.04888   | 0.45133 | 7.28847 |
|                                  | (D * Q)   | 0.02711   | 0.42954 | 7.26683 |
|                                  | (!D * !Q) | 0.05245   | 0.45442 | 7.29354 |
|                                  | (!D * !Q) | 0.03037   | 0.43231 | 7.27144 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_DLAT\_1

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT |     | OUTPUT |
|-------|-----|--------|
| D     | CLK | Q      |
| x     | 0   | IQ     |
| 0     | 1   | 0      |
| 1     | 1   | 1      |

## Footprint

| Cell Name                       | Area     |
|---------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__dlat_1 | 60.48000 |

## Pin Capacitance Information

| Cell Name                       | Pin Cap(pf) |         | Max Cap(pf) |
|---------------------------------|-------------|---------|-------------|
|                                 | D           | CLK     | Q           |
| gf180mcu_osu_sc_gp9t3v3__dlat_1 | 0.00395     | 0.00881 | 15.71026    |

## Leakage Information

| Cell Name                       | Leakage(nW) |         |         |
|---------------------------------|-------------|---------|---------|
|                                 | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__dlat_1 | 0.00000     | 0.00412 | 0.00471 |

## Delay Information

Delay(ns) to Q rising :

| Cell Name                       | Timing Arc(Dir) | Delay(ns) |         |          |
|---------------------------------|-----------------|-----------|---------|----------|
|                                 |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__dlat_1 | CLK->Q (RR)     | 0.21619   | 1.77828 | 61.37890 |
|                                 | D->Q (RR)       | 0.24360   | 1.81150 | 63.36240 |

Delay(ns) to Q falling :

| Cell Name                       | Timing Arc(Dir) | Delay(ns) |         |          |
|---------------------------------|-----------------|-----------|---------|----------|
|                                 |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__dlat_1 | CLK->Q (RF)     | 0.27946   | 1.62693 | 56.69900 |
|                                 | D->Q (FF)       | 0.27134   | 2.43847 | 70.66860 |

## Constraint Information

Constraints(ns) for D rising :

| Cell Name                       | Timing Check | Ref Pin(trans) | Reference Slew Rate(ns) |          |           |
|---------------------------------|--------------|----------------|-------------------------|----------|-----------|
|                                 |              |                | first                   | mid      | last      |
| gf180mcu_osu_sc_gp9t3v3__dlat_1 | hold         | CLK (F)        | -0.13876                | -0.68689 | -20.42280 |
|                                 | setup        | CLK (F)        | 0.14831                 | 1.27522  | 44.30140  |

Constraints(ns) for D falling :

| Cell Name                       | Timing Check | Ref Pin(trans) | Reference Slew Rate(ns) |          |          |
|---------------------------------|--------------|----------------|-------------------------|----------|----------|
|                                 |              |                | first                   | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__dlat_1 | hold         | CLK (F)        | -0.12434                | -0.14893 | 6.94356  |
|                                 | setup        | CLK (F)        | 0.13600                 | 0.15501  | -6.94817 |

Constraints(ns) for CLK rising (conditional):

| Cell Name                       | Timing Check    | Ref Pin(trans) | When | Reference Slew Rate(ns) |         |           |
|---------------------------------|-----------------|----------------|------|-------------------------|---------|-----------|
|                                 |                 |                |      | first                   | mid     | last      |
| gf180mcu_osu_sc_gp9t3v3__dlat_1 | min_pulse_width | CLK ()         | D    | 0.12074                 | 4.36523 | 165.00100 |
|                                 | min_pulse_width | CLK ()         | !D   | 0.14935                 | 4.36523 | 165.00100 |

## Power Information

Internal switching power(pJ) to Q rising :

| Cell Name                       | Input | Power(pJ) |         |         |
|---------------------------------|-------|-----------|---------|---------|
|                                 |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__dlat_1 | CLK   | 0.06712   | 0.64198 | 9.50834 |
|                                 | CLK   | 0.11167   | 0.68657 | 9.52895 |
|                                 | D     | 0.06860   | 0.42816 | 6.80558 |
|                                 | D     | 0.09640   | 0.45554 | 6.81192 |

Internal switching power(pJ) to Q falling :

| Cell Name                       | Input | Power(pJ) |         |         |
|---------------------------------|-------|-----------|---------|---------|
|                                 |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__dlat_1 | CLK   | 0.08993   | 0.46841 | 7.20560 |
|                                 | CLK   | 0.11648   | 0.49519 | 7.23500 |
|                                 | D     | 0.11022   | 0.47014 | 6.80523 |
|                                 | D     | 0.08190   | 0.44272 | 6.77859 |

Passive power(pJ) for D rising (conditional):

| Cell Name                       | When | Power(pJ) |          |          |
|---------------------------------|------|-----------|----------|----------|
|                                 |      | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__dlat_1 | !CLK | -0.01279  | -0.01360 | -0.01343 |
|                                 | !CLK | 0.00703   | 0.00659  | 0.00644  |

Passive power(pJ) for D falling (conditional):

| Cell Name                       | When | Power(pJ) |          |          |
|---------------------------------|------|-----------|----------|----------|
|                                 |      | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__dlat_1 | !CLK | 0.01345   | 0.01366  | 0.01343  |
|                                 | !CLK | -0.00634  | -0.00649 | -0.00642 |

Passive power(pJ) for CLK rising (conditional):

| Cell Name                       | When      | Power(pJ) |         |         |
|---------------------------------|-----------|-----------|---------|---------|
|                                 |           | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__dlat_1 | (D * Q)   | -0.00454  | 0.36753 | 6.85868 |
|                                 | (D * Q)   | 0.02991   | 0.40214 | 6.89326 |
|                                 | (!D * !Q) | -0.00459  | 0.36426 | 6.85764 |
|                                 | (!D * !Q) | 0.03317   | 0.40198 | 6.89543 |

Passive power(pJ) for CLK falling (conditional):

| Cell Name                       | When      | Power(pJ) |         |         |
|---------------------------------|-----------|-----------|---------|---------|
|                                 |           | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__dlat_1 | (D * Q)   | 0.03270   | 0.41016 | 6.89759 |
|                                 | (D * Q)   | -0.00168  | 0.37579 | 6.86298 |
|                                 | (!D * !Q) | 0.03550   | 0.41036 | 6.89811 |
|                                 | (!D * !Q) | -0.00230  | 0.37247 | 6.86018 |



# GF180MCU\_OSU\_SC\_GP9T3V3\_\_INV\_16

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT | OUTPUT |
|-------|--------|
| A     | Y      |
| 0     | 1      |
| 1     | 0      |

## Footprint

| Cell Name                       | Area     |
|---------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__inv_16 | 94.50000 |

## Pin Capacitance Information

| Cell Name                       | Pin Cap(pf) | Max Cap(pf) |
|---------------------------------|-------------|-------------|
|                                 | A           | Y           |
| gf180mcu_osu_sc_gp9t3v3__inv_16 | 0.06475     | 234.89905   |

## Leakage Information

| Cell Name                       | Leakage(nW) |         |         |
|---------------------------------|-------------|---------|---------|
|                                 | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__inv_16 | 0.00000     | 0.01179 | 0.01439 |

## Delay Information

Delay(ns) to Y rising :

| Cell Name                       | Timing Arc(Dir) | Delay(ns) |         |          |
|---------------------------------|-----------------|-----------|---------|----------|
|                                 |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__inv_16 | A->Y (FR)       | 0.03120   | 2.15006 | 98.41110 |

Delay(ns) to Y falling :

| Cell Name                       | Timing Arc(Dir) | Delay(ns) |         |          |
|---------------------------------|-----------------|-----------|---------|----------|
|                                 |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__inv_16 | A->Y (RF)       | 0.02337   | 1.37518 | 83.67180 |

## Power Information

Internal switching power(pJ) to Y rising :

| Cell Name                       | Input | Power(pJ) |         |          |
|---------------------------------|-------|-----------|---------|----------|
|                                 |       | first     | mid     | last     |
| gf180mcu_osu_sc_gp9t3v3__inv_16 | A     | 0.32160   | 4.42993 | 38.24140 |
|                                 | A     | -0.02739  | 4.07467 | 37.86540 |

Internal switching power(pJ) to Y falling :

| Cell Name                       | Input | Power(pJ) |         |          |
|---------------------------------|-------|-----------|---------|----------|
|                                 |       | first     | mid     | last     |
| gf180mcu_osu_sc_gp9t3v3__inv_16 | A     | -0.05392  | 3.89220 | 34.47280 |
|                                 | A     | 0.29638   | 4.24427 | 34.82470 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_INV\_1

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT | OUTPUT |
|-------|--------|
| A     | Y      |
| 0     | 1      |
| 1     | 0      |

## Footprint

| Cell Name                      | Area     |
|--------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__inv_1 | 15.12000 |

## Pin Capacitance Information

| Cell Name                      | Pin Cap(pf) | Max Cap(pf) |
|--------------------------------|-------------|-------------|
|                                | A           | Y           |
| gf180mcu_osu_sc_gp9t3v3__inv_1 | 0.00405     | 14.68244    |

## Leakage Information

| Cell Name                      | Leakage(nW) |         |         |
|--------------------------------|-------------|---------|---------|
|                                | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__inv_1 | 0.00000     | 0.00074 | 0.00090 |

## Delay Information

Delay(ns) to Y rising :

| Cell Name                      | Timing Arc(Dir) | Delay(ns) |         |          |
|--------------------------------|-----------------|-----------|---------|----------|
|                                |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__inv_1 | A->Y (FR)       | 0.03869   | 3.38170 | 98.41570 |

Delay(ns) to Y falling :

| Cell Name                      | Timing Arc(Dir) | Delay(ns) |         |          |
|--------------------------------|-----------------|-----------|---------|----------|
|                                |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__inv_1 | A->Y (RF)       | 0.03096   | 2.61905 | 83.67590 |

## Power Information

Internal switching power(pJ) to Y rising :

| Cell Name                      | Input | Power(pJ) |         |         |
|--------------------------------|-------|-----------|---------|---------|
|                                |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__inv_1 | A     | 0.01992   | 0.22335 | 2.39002 |
|                                | A     | -0.00196  | 0.20134 | 2.36651 |

Internal switching power(pJ) to Y falling :

| Cell Name                      | Input | Power(pJ) |         |         |
|--------------------------------|-------|-----------|---------|---------|
|                                |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__inv_1 | A     | -0.00449  | 0.17814 | 2.15348 |
|                                | A     | 0.01738   | 0.20005 | 2.17547 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_INV\_2

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT | OUTPUT |
|-------|--------|
| A     | Y      |
| 0     | 1      |
| 1     | 0      |

## Footprint

| Cell Name                      | Area     |
|--------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__inv_2 | 20.79000 |

## Pin Capacitance Information

| Cell Name                      | Pin Cap(pf) | Max Cap(pf) |
|--------------------------------|-------------|-------------|
|                                | A           | Y           |
| gf180mcu_osu_sc_gp9t3v3__inv_2 | 0.00808     | 29.36374    |

## Leakage Information

| Cell Name                      | Leakage(nW) |         |         |
|--------------------------------|-------------|---------|---------|
|                                | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__inv_2 | 0.00000     | 0.00147 | 0.00180 |

## Delay Information

Delay(ns) to Y rising :

| Cell Name                      | Timing Arc(Dir) | Delay(ns) |         |          |
|--------------------------------|-----------------|-----------|---------|----------|
|                                |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__inv_2 | A->Y (FR)       | 0.03414   | 2.91144 | 98.41320 |

Delay(ns) to Y falling :

| Cell Name                      | Timing Arc(Dir) | Delay(ns) |         |          |
|--------------------------------|-----------------|-----------|---------|----------|
|                                |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__inv_2 | A->Y (RF)       | 0.02634   | 2.15085 | 83.67370 |



## Power Information

Internal switching power(pJ) to Y rising :

| Cell Name                      | Input | Power(pJ) |         |         |
|--------------------------------|-------|-----------|---------|---------|
|                                |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__inv_2 | A     | 0.03999   | 0.47013 | 4.78011 |
|                                | A     | -0.00363  | 0.42539 | 4.73310 |

Internal switching power(pJ) to Y falling :

| Cell Name                      | Input | Power(pJ) |         |         |
|--------------------------------|-------|-----------|---------|---------|
|                                |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__inv_2 | A     | -0.00881  | 0.39669 | 4.30699 |
|                                | A     | 0.03480   | 0.44042 | 4.35098 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_INV\_4

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT | OUTPUT |
|-------|--------|
| A     | Y      |
| 0     | 1      |
| 1     | 0      |

## Footprint

| Cell Name                      | Area     |
|--------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__inv_4 | 30.24000 |

## Pin Capacitance Information

| Cell Name                      | Pin Cap(pf) | Max Cap(pf) |
|--------------------------------|-------------|-------------|
|                                | A           | Y           |
| gf180mcu_osu_sc_gp9t3v3__inv_4 | 0.01618     | 58.72635    |

## Leakage Information

| Cell Name                      | Leakage(nW) |         |         |
|--------------------------------|-------------|---------|---------|
|                                | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__inv_4 | 0.00000     | 0.00295 | 0.00360 |

## Delay Information

Delay(ns) to Y rising :

| Cell Name                      | Timing Arc(Dir) | Delay(ns) |         |          |
|--------------------------------|-----------------|-----------|---------|----------|
|                                |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__inv_4 | A->Y (FR)       | 0.03168   | 2.52828 | 98.41200 |

Delay(ns) to Y falling :

| Cell Name                      | Timing Arc(Dir) | Delay(ns) |         |          |
|--------------------------------|-----------------|-----------|---------|----------|
|                                |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__inv_4 | A->Y (RF)       | 0.02386   | 1.76026 | 83.67260 |

## Power Information

Internal switching power(pJ) to Y rising :

| Cell Name                      | Input | Power(pJ) |         |         |
|--------------------------------|-------|-----------|---------|---------|
|                                |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__inv_4 | A     | 0.08065   | 1.02183 | 9.56029 |
|                                | A     | -0.00700  | 0.93247 | 9.46628 |

Internal switching power(pJ) to Y falling :

| Cell Name                      | Input | Power(pJ) |         |         |
|--------------------------------|-------|-----------|---------|---------|
|                                |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__inv_4 | A     | -0.01761  | 0.87513 | 8.61403 |
|                                | A     | 0.06989   | 0.96373 | 8.70200 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_INV\_8

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT | OUTPUT |
|-------|--------|
| A     | Y      |
| 0     | 1      |
| 1     | 0      |

## Footprint

| Cell Name                      | Area     |
|--------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__inv_8 | 52.92000 |

## Pin Capacitance Information

| Cell Name                      | Pin Cap(pf) | Max Cap(pf) |
|--------------------------------|-------------|-------------|
|                                | A           | Y           |
| gf180mcu_osu_sc_gp9t3v3__inv_8 | 0.03246     | 117.45156   |

## Leakage Information

| Cell Name                      | Leakage(nW) |         |         |
|--------------------------------|-------------|---------|---------|
|                                | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__inv_8 | 0.00000     | 0.00590 | 0.00720 |

## Delay Information

Delay(ns) to Y rising :

| Cell Name                      | Timing Arc(Dir) | Delay(ns) |         |          |
|--------------------------------|-----------------|-----------|---------|----------|
|                                |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__inv_8 | A->Y (FR)       | 0.03034   | 2.20462 | 98.41140 |

Delay(ns) to Y falling :

| Cell Name                      | Timing Arc(Dir) | Delay(ns) |         |          |
|--------------------------------|-----------------|-----------|---------|----------|
|                                |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__inv_8 | A->Y (RF)       | 0.02253   | 1.43037 | 83.67200 |

## Power Information

Internal switching power(pJ) to Y rising :

| Cell Name                      | Input | Power(pJ) |         |          |
|--------------------------------|-------|-----------|---------|----------|
|                                |       | first     | mid     | last     |
| gf180mcu_osu_sc_gp9t3v3__inv_8 | A     | 0.16151   | 2.18866 | 19.12070 |
|                                | A     | -0.01369  | 2.01180 | 18.93270 |

Internal switching power(pJ) to Y falling :

| Cell Name                      | Input | Power(pJ) |         |          |
|--------------------------------|-------|-----------|---------|----------|
|                                |       | first     | mid     | last     |
| gf180mcu_osu_sc_gp9t3v3__inv_8 | A     | -0.03509  | 1.90803 | 17.22810 |
|                                | A     | 0.14004   | 2.08375 | 17.40400 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_MUX2\_1

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT |   |     | OUTPUT |
|-------|---|-----|--------|
| A     | B | Sel | Y      |
| 0     | 0 | x   | 0      |
| 0     | 1 | 0   | 0      |
| x     | 1 | 1   | 1      |
| 1     | x | 0   | 1      |
| 1     | 0 | 1   | 0      |

## Footprint

| Cell Name                       | Area     |
|---------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__mux2_1 | 32.13000 |

## Pin Capacitance Information

| Cell Name                       | Pin Cap(pf) |         |         | Max Cap(pf) |
|---------------------------------|-------------|---------|---------|-------------|
|                                 | A           | B       | Sel     | Y           |
| gf180mcu_osu_sc_gp9t3v3__mux2_1 | 0.01046     | 0.01046 | 0.00808 | 0.00364     |

## Leakage Information

| Cell Name                       | Leakage(nW) |         |         |
|---------------------------------|-------------|---------|---------|
|                                 | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__mux2_1 | 0.00000     | 0.00197 | 0.00201 |



## Delay Information

Delay(ns) to Y rising (conditional):

| Cell Name                       | Timing Arc(Dir) | When     | Delay(ns) |          |          |
|---------------------------------|-----------------|----------|-----------|----------|----------|
|                                 |                 |          | First     | Mid      | Last     |
| gf180mcu_osu_sc_gp9t3v3__mux2_1 | A->Y (RR)       | -        | 0.01760   | 0.02195  | 0.02576  |
|                                 | B->Y (RR)       | -        | 0.01925   | 0.02200  | 0.02577  |
|                                 | Sel->Y (RR)     | (!A * B) | 0.06266   | -0.04996 | -5.90009 |
|                                 | Sel->Y (FR)     | (A * !B) | 0.04400   | 0.58697  | 7.30364  |

Delay(ns) to Y falling (conditional):

| Cell Name                       | Timing Arc(Dir) | When     | Delay(ns) |          |          |
|---------------------------------|-----------------|----------|-----------|----------|----------|
|                                 |                 |          | First     | Mid      | Last     |
| gf180mcu_osu_sc_gp9t3v3__mux2_1 | A->Y (FF)       | -        | 0.02158   | 0.02204  | 0.02546  |
|                                 | B->Y (FF)       | -        | 0.01939   | 0.02199  | 0.02545  |
|                                 | Sel->Y (FF)     | (!A * B) | 0.07299   | 0.67004  | 7.22989  |
|                                 | Sel->Y (RF)     | (A * !B) | 0.03641   | -0.18075 | -6.49607 |

## Power Information

Internal switching power(pJ) to Y rising (conditional):

| Cell Name                       | Input | When     | Power(pJ) |          |          |
|---------------------------------|-------|----------|-----------|----------|----------|
|                                 |       |          | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__mux2_1 | A     | -        | -0.01951  | -0.01964 | -0.01961 |
|                                 | A     | -        | 0.00678   | 0.00682  | 0.00681  |
|                                 | B     | -        | -0.01287  | -0.01297 | -0.01296 |
|                                 | B     | -        | 0.01753   | 0.01768  | 0.01766  |
|                                 | Sel   | (A * !B) | 0.01688   | 0.39064  | 6.88395  |
|                                 | Sel   | (A * !B) | 0.00283   | 0.37730  | 6.88505  |
|                                 | Sel   | (!A * B) | -0.01420  | 0.35466  | 6.84594  |
|                                 | Sel   | (!A * B) | 0.04382   | 0.41447  | 6.92876  |

Internal switching power(pJ) to Y falling (conditional):

| Cell Name                       | Input | When     | Power(pJ) |          |          |
|---------------------------------|-------|----------|-----------|----------|----------|
|                                 |       |          | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__mux2_1 | A     | -        | 0.01951   | 0.01964  | 0.01962  |
|                                 | A     | -        | -0.00678  | -0.00682 | -0.00681 |
|                                 | B     | -        | 0.01288   | 0.01297  | 0.01296  |
|                                 | B     | -        | -0.01753  | -0.01763 | -0.01761 |
|                                 | Sel   | (A * !B) | 0.00494   | 0.37583  | 6.87146  |
|                                 | Sel   | (A * !B) | 0.01900   | 0.39097  | 6.90068  |
|                                 | Sel   | (!A * B) | 0.05063   | 0.42021  | 6.91155  |
|                                 | Sel   | (!A * B) | -0.00736  | 0.36327  | 6.87814  |

Passive power(pJ) for A rising (conditional):

| Cell Name                       | When                            | Power(pJ) |          |          |
|---------------------------------|---------------------------------|-----------|----------|----------|
|                                 |                                 | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__mux2_1 | (B * Sel * Y) + (!B * Sel * !Y) | -0.00347  | -0.00350 | -0.00348 |
|                                 | (B * Sel * Y) + (!B * Sel * !Y) | 0.00262   | 0.00265  | 0.00263  |

Passive power(pJ) for A falling (conditional):

| Cell Name                       | When                              | Power(pJ) |          |          |
|---------------------------------|-----------------------------------|-----------|----------|----------|
|                                 |                                   | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__mux2_1 | $(B * Sel * Y) + (!B * Sel * !Y)$ | 0.00350   | 0.00350  | 0.00348  |
|                                 | $(B * Sel * Y) + (!B * Sel * !Y)$ | -0.00262  | -0.00265 | -0.00263 |

Passive power(pJ) for B rising (conditional):

| Cell Name                       | When                                | Power(pJ) |          |          |
|---------------------------------|-------------------------------------|-----------|----------|----------|
|                                 |                                     | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__mux2_1 | $(A * !Sel * Y) + (!A * !Sel * !Y)$ | -0.00476  | -0.00480 | -0.00476 |
|                                 | $(A * !Sel * Y) + (!A * !Sel * !Y)$ | 0.00200   | 0.00201  | 0.00200  |

Passive power(pJ) for B falling (conditional):

| Cell Name                       | When                                | Power(pJ) |          |          |
|---------------------------------|-------------------------------------|-----------|----------|----------|
|                                 |                                     | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__mux2_1 | $(A * !Sel * Y) + (!A * !Sel * !Y)$ | 0.00476   | 0.00480  | 0.00476  |
|                                 | $(A * !Sel * Y) + (!A * !Sel * !Y)$ | -0.00200  | -0.00201 | -0.00200 |

Passive power(pJ) for Sel rising (conditional):

| Cell Name                       | When             | Power(pJ) |         |         |
|---------------------------------|------------------|-----------|---------|---------|
|                                 |                  | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__mux2_1 | $(A * B * Y)$    | -0.00475  | 0.36733 | 6.85753 |
|                                 | $(A * B * Y)$    | 0.03304   | 0.40533 | 6.89548 |
|                                 | $(!A * !B * !Y)$ | -0.00464  | 0.36737 | 6.85859 |
|                                 | $(!A * !B * !Y)$ | 0.02955   | 0.40205 | 6.89303 |

Passive power(pJ) for Sel falling (conditional):

| Cell Name                       | When           | Power(pJ) |         |         |
|---------------------------------|----------------|-----------|---------|---------|
|                                 |                | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__mux2_1 | (A * B * Y)    | 0.03548   | 0.41120 | 6.89791 |
|                                 | (A * B * Y)    | -0.00237  | 0.37333 | 6.85986 |
|                                 | (!A * !B * !Y) | 0.03220   | 0.40970 | 6.89748 |
|                                 | (!A * !B * !Y) | -0.00204  | 0.37536 | 6.86291 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_NAND2\_1

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT |   | OUTPUT |
|-------|---|--------|
| A     | B | Y      |
| 0     | x | 1      |
| 1     | 0 | 1      |
| 1     | 1 | 0      |

## Footprint

| Cell Name                        | Area     |
|----------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__nand2_1 | 20.79000 |

## Pin Capacitance Information

| Cell Name                        | Pin Cap(pf) |         | Max Cap(pf) |
|----------------------------------|-------------|---------|-------------|
|                                  | A           | B       | Y           |
| gf180mcu_osu_sc_gp9t3v3__nand2_1 | 0.00404     | 0.00402 | 10.44023    |

## Leakage Information

| Cell Name                        | Leakage(nW) |         |         |
|----------------------------------|-------------|---------|---------|
|                                  | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__nand2_1 | 0.00000     | 0.00078 | 0.00115 |

## Delay Information

Delay(ns) to Y rising :

| Cell Name                        | Timing Arc(Dir) | Delay(ns) |         |          |
|----------------------------------|-----------------|-----------|---------|----------|
|                                  |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__nand2_1 | A->Y (FR)       | 0.04424   | 2.93592 | 79.20290 |
|                                  | B->Y (FR)       | 0.05413   | 2.98386 | 79.50170 |

Delay(ns) to Y falling :

| Cell Name                        | Timing Arc(Dir) | Delay(ns) |         |          |
|----------------------------------|-----------------|-----------|---------|----------|
|                                  |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__nand2_1 | A->Y (RF)       | 0.05001   | 3.11731 | 89.80080 |
|                                  | B->Y (RF)       | 0.05094   | 2.53124 | 78.16810 |

## Power Information

Internal switching power(pJ) to Y rising :

| Cell Name                        | Input | Power(pJ) |         |         |
|----------------------------------|-------|-----------|---------|---------|
|                                  |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__nand2_1 | A     | 0.02134   | 0.18792 | 2.19715 |
|                                  | A     | -0.00176  | 0.16380 | 2.15837 |
|                                  | B     | 0.02976   | 0.21189 | 2.38040 |
|                                  | B     | 0.00156   | 0.18303 | 2.33677 |

Internal switching power(pJ) to Y falling :

| Cell Name                        | Input | Power(pJ) |         |         |
|----------------------------------|-------|-----------|---------|---------|
|                                  |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__nand2_1 | A     | -0.00169  | 0.15915 | 2.10089 |
|                                  | A     | 0.02145   | 0.18259 | 2.12769 |
|                                  | B     | -0.00270  | 0.17394 | 2.37665 |
|                                  | B     | 0.02551   | 0.20324 | 2.41500 |

Passive power(pJ) for A rising (conditional):

| Cell Name                        | When     | Power(pJ) |          |          |
|----------------------------------|----------|-----------|----------|----------|
|                                  |          | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__nand2_1 | (!B * Y) | -0.01402  | -0.01399 | -0.01410 |
|                                  | (!B * Y) | 0.00181   | 0.00181  | 0.00172  |

Passive power(pJ) for A falling (conditional):

| Cell Name                        | When     | Power(pJ) |          |          |
|----------------------------------|----------|-----------|----------|----------|
|                                  |          | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__nand2_1 | (!B * Y) | 0.01418   | 0.01426  | 0.01420  |
|                                  | (!B * Y) | -0.00167  | -0.00168 | -0.00167 |

Passive power(pJ) for B rising (conditional):

| Cell Name                        | When     | Power(pJ) |          |          |
|----------------------------------|----------|-----------|----------|----------|
|                                  |          | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__nand2_1 | (!A * Y) | -0.01349  | -0.01357 | -0.01353 |
|                                  | (!A * Y) | 0.00646   | 0.00656  | 0.00648  |

Passive power(pJ) for B falling (conditional):

| Cell Name                        | When     | Power(pJ) |          |          |
|----------------------------------|----------|-----------|----------|----------|
|                                  |          | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__nand2_1 | (!A * Y) | 0.01362   | 0.01388  | 0.01357  |
|                                  | (!A * Y) | -0.00630  | -0.00652 | -0.00646 |



# GF180MCU\_OSU\_SC\_GP9T3V3\_\_NOR2\_1

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT |   | OUTPUT |
|-------|---|--------|
| A     | B | Y      |
| 0     | 0 | 1      |
| x     | 1 | 0      |
| 1     | x | 0      |

## Footprint

| Cell Name                       | Area     |
|---------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__nor2_1 | 20.79000 |

## Pin Capacitance Information

| Cell Name                       | Pin Cap(pf) |         | Max Cap(pf) |
|---------------------------------|-------------|---------|-------------|
|                                 | A           | B       | Y           |
| gf180mcu_osu_sc_gp9t3v3__nor2_1 | 0.00399     | 0.00405 | 7.76937     |

## Leakage Information

| Cell Name                       | Leakage(nW) |         |         |
|---------------------------------|-------------|---------|---------|
|                                 | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__nor2_1 | 0.00000     | 0.00082 | 0.00180 |

## Delay Information

Delay(ns) to Y rising :

| Cell Name                       | Timing Arc(Dir) | Delay(ns) |         |          |
|---------------------------------|-----------------|-----------|---------|----------|
|                                 |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__nor2_1 | A->Y (FR)       | 0.07295   | 3.29510 | 86.27430 |
|                                 | B->Y (FR)       | 0.05957   | 3.90394 | 97.81030 |

Delay(ns) to Y falling :

| Cell Name                       | Timing Arc(Dir) | Delay(ns) |         |          |
|---------------------------------|-----------------|-----------|---------|----------|
|                                 |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__nor2_1 | A->Y (RF)       | 0.04977   | 1.94863 | 53.01870 |
|                                 | B->Y (RF)       | 0.03596   | 1.86784 | 52.46390 |

## Power Information

Internal switching power(pJ) to Y rising :

| Cell Name                       | Input | Power(pJ) |         |         |
|---------------------------------|-------|-----------|---------|---------|
|                                 |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__nor2_1 | A     | 0.03017   | 0.22817 | 2.99726 |
|                                 | A     | -0.00171  | 0.19537 | 2.96359 |
|                                 | B     | 0.02161   | 0.19103 | 2.46298 |
|                                 | B     | -0.00083  | 0.16792 | 2.44022 |

Internal switching power(pJ) to Y falling :

| Cell Name                       | Input | Power(pJ) |         |         |
|---------------------------------|-------|-----------|---------|---------|
|                                 |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__nor2_1 | A     | 0.00242   | 0.17939 | 2.49191 |
|                                 | A     | 0.03427   | 0.21149 | 2.53682 |
|                                 | B     | -0.00332  | 0.15182 | 2.20967 |
|                                 | B     | 0.01920   | 0.17481 | 2.26171 |

Passive power(pJ) for A rising (conditional):

| Cell Name                       | When     | Power(pJ) |          |          |
|---------------------------------|----------|-----------|----------|----------|
|                                 |          | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__nor2_1 | (B * !Y) | -0.01274  | -0.01340 | -0.01329 |
|                                 | (B * !Y) | 0.00712   | 0.00655  | 0.00652  |

Passive power(pJ) for A falling (conditional):

| Cell Name                       | When     | Power(pJ) |          |          |
|---------------------------------|----------|-----------|----------|----------|
|                                 |          | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__nor2_1 | (B * !Y) | 0.01338   | 0.01340  | 0.01329  |
|                                 | (B * !Y) | -0.00650  | -0.00649 | -0.00648 |

Passive power(pJ) for B rising (conditional):

| Cell Name                       | When     | Power(pJ) |          |          |
|---------------------------------|----------|-----------|----------|----------|
|                                 |          | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__nor2_1 | (A * !Y) | -0.00434  | -0.00445 | -0.00434 |
|                                 | (A * !Y) | 0.00778   | 0.00800  | 0.00781  |

Passive power(pJ) for B falling (conditional):

| Cell Name                       | When     | Power(pJ) |          |          |
|---------------------------------|----------|-----------|----------|----------|
|                                 |          | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__nor2_1 | (A * !Y) | 0.00470   | 0.00467  | 0.00448  |
|                                 | (A * !Y) | -0.00745  | -0.00741 | -0.00767 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_OAI21\_1

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT |    |   | OUTPUT |
|-------|----|---|--------|
| A0    | A1 | B | Y      |
| 0     | 0  | x | 1      |
| x     | 1  | 0 | 1      |
| x     | 1  | 1 | 0      |
| 1     | x  | 0 | 1      |
| 1     | x  | 1 | 0      |

## Footprint

| Cell Name                        | Area     |
|----------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__oai21_1 | 26.46000 |

## Pin Capacitance Information

| Cell Name                        | Pin Cap(pf) |         |         | Max Cap(pf) |
|----------------------------------|-------------|---------|---------|-------------|
|                                  | A0          | A1      | B       | Y           |
| gf180mcu_osu_sc_gp9t3v3__oai21_1 | 0.00396     | 0.00402 | 0.00404 | 7.88413     |

## Leakage Information

| Cell Name                        | Leakage(nW) |         |         |
|----------------------------------|-------------|---------|---------|
|                                  | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__oai21_1 | 0.00000     | 0.00096 | 0.00152 |

## Delay Information

Delay(ns) to Y rising :

| Cell Name                        | Timing Arc(Dir) | Delay(ns) |         |          |
|----------------------------------|-----------------|-----------|---------|----------|
|                                  |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__oai21_1 | A0->Y (FR)      | 0.09873   | 3.25262 | 85.78900 |
|                                  | A1->Y (FR)      | 0.08302   | 3.87867 | 97.57670 |
|                                  | B->Y (FR)       | 0.04391   | 2.71884 | 67.76420 |

Delay(ns) to Y falling :

| Cell Name                        | Timing Arc(Dir) | Delay(ns) |         |          |
|----------------------------------|-----------------|-----------|---------|----------|
|                                  |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__oai21_1 | A0->Y (RF)      | 0.07776   | 2.18146 | 61.12010 |
|                                  | A1->Y (RF)      | 0.05671   | 2.10161 | 60.50820 |
|                                  | B->Y (RF)       | 0.07413   | 2.82563 | 74.10740 |

## Power Information

Internal switching power(pJ) to Y rising :

| Cell Name                        | Input | Power(pJ) |         |         |
|----------------------------------|-------|-----------|---------|---------|
|                                  |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__oai21_1 | A0    | 0.04061   | 0.21107 | 2.51348 |
|                                  | A0    | 0.00240   | 0.17243 | 2.47536 |
|                                  | A1    | 0.03138   | 0.17853 | 2.06542 |
|                                  | A1    | 0.00268   | 0.14919 | 2.03669 |
|                                  | B     | 0.02126   | 0.21902 | 2.83693 |
|                                  | B     | -0.00192  | 0.19583 | 2.80625 |

Internal switching power(pJ) to Y falling :

| Cell Name                        | Input | Power(pJ) |         |         |
|----------------------------------|-------|-----------|---------|---------|
|                                  |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__oai21_1 | A0    | 0.00496   | 0.16308 | 2.26661 |
|                                  | A0    | 0.04315   | 0.20163 | 2.30432 |
|                                  | A1    | -0.00155  | 0.13868 | 2.02228 |
|                                  | A1    | 0.02719   | 0.16783 | 2.05100 |
|                                  | B     | -0.00143  | 0.18897 | 2.68954 |
|                                  | B     | 0.02174   | 0.21220 | 2.71307 |

Passive power(pJ) for A0 rising (conditional):

| Cell Name                        | When           | Power(pJ) |          |          |
|----------------------------------|----------------|-----------|----------|----------|
|                                  |                | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__oai21_1 | (A1 * B * !Y)  | -0.01268  | -0.01341 | -0.01329 |
|                                  | (A1 * B * !Y)  | 0.00717   | 0.00655  | 0.00652  |
|                                  | (A1 * !B * Y)  | -0.01327  | -0.01339 | -0.01328 |
|                                  | (A1 * !B * Y)  | 0.00658   | 0.00655  | 0.00652  |
|                                  | (!A1 * !B * Y) | -0.01353  | -0.01355 | -0.01353 |
|                                  | (!A1 * !B * Y) | 0.00650   | 0.00654  | 0.00646  |

Passive power(pJ) for A0 falling (conditional):

| Cell Name                        | When           | Power(pJ) |          |          |
|----------------------------------|----------------|-----------|----------|----------|
|                                  |                | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__oai21_1 | (A1 * B * !Y)  | 0.01331   | 0.01341  | 0.01329  |
|                                  | (A1 * B * !Y)  | -0.00648  | -0.00648 | -0.00647 |
|                                  | (A1 * !B * Y)  | 0.01328   | 0.01339  | 0.01328  |
|                                  | (A1 * !B * Y)  | -0.00649  | -0.00650 | -0.00649 |
|                                  | (!A1 * !B * Y) | 0.01360   | 0.01355  | 0.01357  |
|                                  | (!A1 * !B * Y) | -0.00633  | -0.00648 | -0.00646 |

Passive power(pJ) for A1 rising (conditional):

| Cell Name                        | When          | Power(pJ) |          |          |
|----------------------------------|---------------|-----------|----------|----------|
|                                  |               | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__oai21_1 | (A0 * B * !Y) | -0.00435  | -0.00444 | -0.00434 |
|                                  | (A0 * B * !Y) | 0.00781   | 0.00800  | 0.00781  |
|                                  | (!B * Y)      | -0.01328  | -0.01341 | -0.01326 |
|                                  | (!B * Y)      | 0.00657   | 0.00661  | 0.00652  |

Passive power(pJ) for A1 falling (conditional):

| Cell Name                        | When          | Power(pJ) |          |          |
|----------------------------------|---------------|-----------|----------|----------|
|                                  |               | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__oai21_1 | (A0 * B * !Y) | 0.00465   | 0.00468  | 0.00448  |
|                                  | (A0 * B * !Y) | -0.00737  | -0.00741 | -0.00767 |
|                                  | (!B * Y)      | 0.01331   | 0.01341  | 0.01326  |
|                                  | (!B * Y)      | -0.00649  | -0.00649 | -0.00649 |

Passive power(pJ) for B rising (conditional):

| Cell Name                        | When            | Power(pJ) |          |          |
|----------------------------------|-----------------|-----------|----------|----------|
|                                  |                 | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__oai21_1 | (!A0 * !A1 * Y) | -0.01394  | -0.01392 | -0.01408 |
|                                  | (!A0 * !A1 * Y) | 0.00187   | 0.00187  | 0.00174  |



**Passive power(pJ) for B falling (conditional):**

| Cell Name                        | When            | Power(pJ) |          |          |
|----------------------------------|-----------------|-----------|----------|----------|
|                                  |                 | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__oai21_1 | (!A0 * !A1 * Y) | 0.01420   | 0.01424  | 0.01420  |
|                                  | (!A0 * !A1 * Y) | -0.00168  | -0.00168 | -0.00167 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_OAI22\_1

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT |    |    |    | OUTPUT |
|-------|----|----|----|--------|
| A0    | A1 | B0 | B1 | Y      |
| 0     | 0  | x  | x  | 1      |
| x     | 1  | 0  | 0  | 1      |
| x     | 1  | x  | 1  | 0      |
| x     | 1  | 1  | x  | 0      |
| 1     | x  | 0  | 0  | 1      |
| 1     | x  | x  | 1  | 0      |
| 1     | x  | 1  | x  | 0      |

## Footprint

| Cell Name                        | Area     |
|----------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__oai22_1 | 35.91000 |

## Pin Capacitance Information

| Cell Name                        | Pin Cap(pf) |         |         |         | Max Cap(pf) |
|----------------------------------|-------------|---------|---------|---------|-------------|
|                                  | A0          | A1      | B0      | B1      | Y           |
| gf180mcu_osu_sc_gp9t3v3__oai22_1 | 0.00396     | 0.00403 | 0.00407 | 0.00399 | 7.84087     |

## Leakage Information

| Cell Name                        | Leakage(nW) |         |         |
|----------------------------------|-------------|---------|---------|
|                                  | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__oai22_1 | 0.00000     | 0.00125 | 0.00180 |

## Delay Information

Delay(ns) to Y rising :

| Cell Name                        | Timing Arc(Dir) | Delay(ns) |         |          |
|----------------------------------|-----------------|-----------|---------|----------|
|                                  |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__oai22_1 | A0->Y (FR)      | 0.11937   | 3.30864 | 86.10960 |
|                                  | A1->Y (FR)      | 0.10340   | 3.92366 | 97.77950 |
|                                  | B0->Y (FR)      | 0.06684   | 3.86402 | 97.51380 |
|                                  | B1->Y (FR)      | 0.08096   | 3.23971 | 85.71610 |

Delay(ns) to Y falling :

| Cell Name                        | Timing Arc(Dir) | Delay(ns) |         |          |
|----------------------------------|-----------------|-----------|---------|----------|
|                                  |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__oai22_1 | A0->Y (RF)      | 0.11041   | 2.21264 | 60.88040 |
|                                  | A1->Y (RF)      | 0.08792   | 2.13399 | 60.27070 |
|                                  | B0->Y (RF)      | 0.08011   | 2.70094 | 72.32760 |
|                                  | B1->Y (RF)      | 0.10167   | 2.77794 | 72.78850 |

## Power Information

Internal switching power(pJ) to Y rising :

| Cell Name                        | Input | Power(pJ) |         |         |
|----------------------------------|-------|-----------|---------|---------|
|                                  |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__oai22_1 | A0    | 0.05040   | 0.23239 | 2.74572 |
|                                  | A0    | 0.00714   | 0.18849 | 2.70089 |
|                                  | A1    | 0.04116   | 0.19712 | 2.25632 |
|                                  | A1    | 0.00735   | 0.16284 | 2.22231 |
|                                  | B0    | 0.02308   | 0.17441 | 2.16357 |
|                                  | B0    | -0.00065  | 0.14994 | 2.13974 |
|                                  | B1    | 0.03184   | 0.20660 | 2.60981 |
|                                  | B1    | -0.00138  | 0.17245 | 2.57660 |

Internal switching power(pJ) to Y falling :

| Cell Name                        | Input | Power(pJ) |         |         |
|----------------------------------|-------|-----------|---------|---------|
|                                  |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__oai22_1 | A0    | 0.00496   | 0.16467 | 2.30038 |
|                                  | A0    | 0.04827   | 0.20836 | 2.35606 |
|                                  | A1    | -0.00157  | 0.13956 | 2.05125 |
|                                  | A1    | 0.03239   | 0.17378 | 2.09821 |
|                                  | B0    | -0.00016  | 0.14143 | 1.99193 |
|                                  | B0    | 0.02365   | 0.16520 | 2.02516 |
|                                  | B1    | 0.00568   | 0.16444 | 2.21341 |
|                                  | B1    | 0.03882   | 0.19825 | 2.25310 |

Passive power(pJ) for A0 rising (conditional):

| Cell Name                        | When                  | Power(pJ) |          |          |
|----------------------------------|-----------------------|-----------|----------|----------|
|                                  |                       | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__oai22_1 | (A1 * B0 * !Y)        | -0.01265  | -0.01352 | -0.01327 |
|                                  | (A1 * B0 * !Y)        | 0.00717   | 0.00655  | 0.00652  |
|                                  | (A1 * !B0 * B1 * !Y)  | -0.01266  | -0.01343 | -0.01328 |
|                                  | (A1 * !B0 * B1 * !Y)  | 0.00717   | 0.00655  | 0.00652  |
|                                  | (A1 * !B0 * !B1 * Y)  | -0.01326  | -0.01339 | -0.01328 |
|                                  | (A1 * !B0 * !B1 * Y)  | 0.00658   | 0.00655  | 0.00652  |
|                                  | (!A1 * !B0 * !B1 * Y) | -0.01354  | -0.01358 | -0.01351 |
|                                  | (!A1 * !B0 * !B1 * Y) | 0.00647   | 0.00655  | 0.00643  |

Passive power(pJ) for A0 falling (conditional):

| Cell Name                        | When                  | Power(pJ) |          |          |
|----------------------------------|-----------------------|-----------|----------|----------|
|                                  |                       | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__oai22_1 | (A1 * B0 * !Y)        | 0.01338   | 0.01352  | 0.01327  |
|                                  | (A1 * B0 * !Y)        | -0.00649  | -0.00648 | -0.00647 |
|                                  | (A1 * !B0 * B1 * !Y)  | 0.01330   | 0.01343  | 0.01328  |
|                                  | (A1 * !B0 * B1 * !Y)  | -0.00648  | -0.00647 | -0.00647 |
|                                  | (A1 * !B0 * !B1 * Y)  | 0.01338   | 0.01339  | 0.01328  |
|                                  | (A1 * !B0 * !B1 * Y)  | -0.00651  | -0.00650 | -0.00649 |
|                                  | (!A1 * !B0 * !B1 * Y) | 0.01369   | 0.01359  | 0.01357  |
|                                  | (!A1 * !B0 * !B1 * Y) | -0.00627  | -0.00646 | -0.00643 |

Passive power(pJ) for A1 rising (conditional):

| Cell Name                        | When                   | Power(pJ) |          |          |
|----------------------------------|------------------------|-----------|----------|----------|
|                                  |                        | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__oai22_1 | $(A0 * B0 * !Y)$       | -0.00443  | -0.00445 | -0.00434 |
|                                  | $(A0 * B0 * !Y)$       | 0.00788   | 0.00800  | 0.00781  |
|                                  | $(A0 * !B0 * B1 * !Y)$ | -0.00433  | -0.00444 | -0.00434 |
|                                  | $(A0 * !B0 * B1 * !Y)$ | 0.00778   | 0.00800  | 0.00781  |
|                                  | $(!B0 * !B1 * Y)$      | -0.01317  | -0.01333 | -0.01317 |
|                                  | $(!B0 * !B1 * Y)$      | 0.00656   | 0.00652  | 0.00652  |

Passive power(pJ) for A1 falling (conditional):

| Cell Name                        | When                   | Power(pJ) |          |          |
|----------------------------------|------------------------|-----------|----------|----------|
|                                  |                        | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__oai22_1 | $(A0 * B0 * !Y)$       | 0.00469   | 0.00468  | 0.00448  |
|                                  | $(A0 * B0 * !Y)$       | -0.00740  | -0.00741 | -0.00767 |
|                                  | $(A0 * !B0 * B1 * !Y)$ | 0.00464   | 0.00468  | 0.00448  |
|                                  | $(A0 * !B0 * B1 * !Y)$ | -0.00737  | -0.00740 | -0.00767 |
|                                  | $(!B0 * !B1 * Y)$      | 0.01317   | 0.01333  | 0.01317  |
|                                  | $(!B0 * !B1 * Y)$      | -0.00649  | -0.00649 | -0.00649 |

Passive power(pJ) for B0 rising (conditional):

| Cell Name                        | When                   | Power(pJ) |          |          |
|----------------------------------|------------------------|-----------|----------|----------|
|                                  |                        | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__oai22_1 | $(A1 * B1 * !Y)$       | -0.00438  | -0.00445 | -0.00434 |
|                                  | $(A1 * B1 * !Y)$       | 0.00780   | 0.00800  | 0.00781  |
|                                  | $(A0 * !A1 * B1 * !Y)$ | -0.00438  | -0.00445 | -0.00434 |
|                                  | $(A0 * !A1 * B1 * !Y)$ | 0.00780   | 0.00799  | 0.00781  |
|                                  | $(!A0 * !A1 * Y)$      | -0.01387  | -0.01411 | -0.01399 |
|                                  | $(!A0 * !A1 * Y)$      | 0.00163   | 0.00164  | 0.00163  |

Passive power(pJ) for B0 falling (conditional):

| Cell Name                        | When                 | Power(pJ) |          |          |
|----------------------------------|----------------------|-----------|----------|----------|
|                                  |                      | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__oai22_1 | (A1 * B1 * !Y)       | 0.00465   | 0.00468  | 0.00448  |
|                                  | (A1 * B1 * !Y)       | -0.00738  | -0.00741 | -0.00767 |
|                                  | (A0 * !A1 * B1 * !Y) | 0.00464   | 0.00468  | 0.00449  |
|                                  | (A0 * !A1 * B1 * !Y) | -0.00738  | -0.00741 | -0.00766 |
|                                  | (!A0 * !A1 * Y)      | 0.01408   | 0.01430  | 0.01399  |
|                                  | (!A0 * !A1 * Y)      | -0.00144  | -0.00146 | -0.00153 |

Passive power(pJ) for B1 rising (conditional):

| Cell Name                        | When                 | Power(pJ) |          |          |
|----------------------------------|----------------------|-----------|----------|----------|
|                                  |                      | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__oai22_1 | (A1 * B0 * !Y)       | -0.01269  | -0.01342 | -0.01328 |
|                                  | (A1 * B0 * !Y)       | 0.00713   | 0.00654  | 0.00652  |
|                                  | (A0 * !A1 * B0 * !Y) | -0.01267  | -0.01341 | -0.01332 |
|                                  | (A0 * !A1 * B0 * !Y) | 0.00714   | 0.00654  | 0.00652  |
|                                  | (!A0 * !A1 * Y)      | -0.01388  | -0.01422 | -0.01406 |
|                                  | (!A0 * !A1 * Y)      | 0.00163   | 0.00166  | 0.00163  |

Passive power(pJ) for B1 falling (conditional):

| Cell Name                        | When                 | Power(pJ) |          |          |
|----------------------------------|----------------------|-----------|----------|----------|
|                                  |                      | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__oai22_1 | (A1 * B0 * !Y)       | 0.01332   | 0.01342  | 0.01328  |
|                                  | (A1 * B0 * !Y)       | -0.00648  | -0.00647 | -0.00647 |
|                                  | (A0 * !A1 * B0 * !Y) | 0.01340   | 0.01341  | 0.01332  |
|                                  | (A0 * !A1 * B0 * !Y) | -0.00650  | -0.00649 | -0.00648 |
|                                  | (!A0 * !A1 * Y)      | 0.01434   | 0.01431  | 0.01412  |
|                                  | (!A0 * !A1 * Y)      | -0.00145  | -0.00145 | -0.00153 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_OAI31\_1

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT |    |    |   | OUTPUT |
|-------|----|----|---|--------|
| A0    | A1 | A2 | B | Y      |
| 0     | 0  | 0  | x | 1      |
| 0     | x  | 1  | 0 | 1      |
| 0     | x  | 1  | 1 | 0      |
| x     | 1  | x  | 0 | 1      |
| x     | 1  | x  | 1 | 0      |
| 1     | x  | x  | 0 | 1      |
| 1     | x  | x  | 1 | 0      |

## Footprint

| Cell Name                        | Area     |
|----------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__oai31_1 | 32.13000 |

## Pin Capacitance Information

| Cell Name                        | Pin Cap(pf) |         |         |         | Max Cap(pf) |
|----------------------------------|-------------|---------|---------|---------|-------------|
|                                  | A0          | A1      | A2      | B       | Y           |
| gf180mcu_osu_sc_gp9t3v3__oai31_1 | 0.00395     | 0.00403 | 0.00396 | 0.00404 | 5.43109     |

## Leakage Information

| Cell Name                        | Leakage(nW) |         |         |
|----------------------------------|-------------|---------|---------|
|                                  | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__oai31_1 | 0.00000     | 0.00101 | 0.00216 |



## Delay Information

Delay(ns) to Y rising :

| Cell Name                        | Timing Arc(Dir) | Delay(ns) |         |          |
|----------------------------------|-----------------|-----------|---------|----------|
|                                  |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__oai31_1 | A0->Y (FR)      | 0.14839   | 3.85604 | 90.16140 |
|                                  | A1->Y (FR)      | 0.10795   | 4.28698 | 98.60270 |
|                                  | A2->Y (FR)      | 0.16624   | 3.45910 | 82.47430 |
|                                  | B->Y (FR)       | 0.04382   | 2.43372 | 55.17650 |

Delay(ns) to Y falling :

| Cell Name                        | Timing Arc(Dir) | Delay(ns) |         |          |
|----------------------------------|-----------------|-----------|---------|----------|
|                                  |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__oai31_1 | A0->Y (RF)      | 0.08378   | 1.72214 | 43.54280 |
|                                  | A1->Y (RF)      | 0.06072   | 1.65230 | 42.98220 |
|                                  | A2->Y (RF)      | 0.09118   | 1.80042 | 44.27920 |
|                                  | B->Y (RF)       | 0.08503   | 2.47936 | 57.61430 |

## Power Information

Internal switching power(pJ) to Y rising :

| Cell Name                        | Input | Power(pJ) |         |         |
|----------------------------------|-------|-----------|---------|---------|
|                                  |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__oai31_1 | A0    | 0.04279   | 0.18437 | 2.32376 |
|                                  | A0    | 0.00412   | 0.14556 | 2.28429 |
|                                  | A1    | 0.03344   | 0.16542 | 2.04358 |
|                                  | A1    | 0.00422   | 0.13584 | 2.01398 |
|                                  | A2    | 0.05218   | 0.22332 | 2.92659 |
|                                  | A2    | 0.00418   | 0.17512 | 2.87695 |
|                                  | B     | 0.02121   | 0.24265 | 3.46862 |
|                                  | B     | -0.00198  | 0.21919 | 3.43092 |

Internal switching power(pJ) to Y falling :

| Cell Name                        | Input | Power(pJ) |         |         |
|----------------------------------|-------|-----------|---------|---------|
|                                  |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__oai31_1 | A0    | 0.00647   | 0.13943 | 2.13252 |
|                                  | A0    | 0.04514   | 0.17837 | 2.18426 |
|                                  | A1    | -0.00103  | 0.12079 | 1.92571 |
|                                  | A1    | 0.02823   | 0.15015 | 1.96762 |
|                                  | A2    | 0.01252   | 0.16245 | 2.39546 |
|                                  | A2    | 0.06040   | 0.21060 | 2.45671 |
|                                  | B     | -0.00134  | 0.21094 | 3.29681 |
|                                  | B     | 0.02185   | 0.23435 | 3.33324 |

Passive power(pJ) for A0 rising (conditional):

| Cell Name                        | When                                  | Power(pJ) |          |          |
|----------------------------------|---------------------------------------|-----------|----------|----------|
|                                  |                                       | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__oai31_1 | $(A1 * B * !Y) + (!A1 * A2 * B * !Y)$ | -0.00696  | -0.00698 | -0.00695 |
|                                  | $(A1 * B * !Y) + (!A1 * A2 * B * !Y)$ | 0.00734   | 0.00732  | 0.00686  |
|                                  | $(A1 * !B * Y)$                       | -0.00788  | -0.00786 | -0.00779 |
|                                  | $(A1 * !B * Y)$                       | 0.00654   | 0.00655  | 0.00652  |
|                                  | $(!A1 * !B * Y)$                      | -0.01319  | -0.01346 | -0.01319 |
|                                  | $(!A1 * !B * Y)$                      | 0.00657   | 0.00652  | 0.00652  |

Passive power(pJ) for A0 falling (conditional):

| Cell Name                        | When                                  | Power(pJ) |          |          |
|----------------------------------|---------------------------------------|-----------|----------|----------|
|                                  |                                       | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__oai31_1 | $(A1 * B * !Y) + (!A1 * A2 * B * !Y)$ | 0.00696   | 0.00698  | 0.00695  |
|                                  | $(A1 * B * !Y) + (!A1 * A2 * B * !Y)$ | -0.00649  | -0.00652 | -0.00649 |
|                                  | $(A1 * !B * Y)$                       | 0.00788   | 0.00786  | 0.00779  |
|                                  | $(A1 * !B * Y)$                       | -0.00647  | -0.00649 | -0.00649 |
|                                  | $(!A1 * !B * Y)$                      | 0.01319   | 0.01346  | 0.01319  |
|                                  | $(!A1 * !B * Y)$                      | -0.00648  | -0.00650 | -0.00649 |

Passive power(pJ) for A1 rising (conditional):

| Cell Name                        | When                | Power(pJ) |          |          |
|----------------------------------|---------------------|-----------|----------|----------|
|                                  |                     | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__oai31_1 | (A0 * B * !Y)       | -0.00436  | -0.00444 | -0.00434 |
|                                  | (A0 * B * !Y)       | 0.00781   | 0.00800  | 0.00781  |
|                                  | (A0 * !B * Y)       | -0.01308  | -0.01346 | -0.01323 |
|                                  | (A0 * !B * Y)       | 0.00655   | 0.00661  | 0.00652  |
|                                  | (!A0 * A2 * B * !Y) | -0.00427  | -0.00432 | -0.00425 |
|                                  | (!A0 * A2 * B * !Y) | 0.00781   | 0.00794  | 0.00780  |
|                                  | (!A0 * !B * Y)      | -0.01196  | -0.01267 | -0.01256 |
|                                  | (!A0 * !B * Y)      | 0.00660   | 0.00657  | 0.00652  |

Passive power(pJ) for A1 falling (conditional):

| Cell Name                        | When                | Power(pJ) |          |          |
|----------------------------------|---------------------|-----------|----------|----------|
|                                  |                     | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__oai31_1 | (A0 * B * !Y)       | 0.00461   | 0.00468  | 0.00448  |
|                                  | (A0 * B * !Y)       | -0.00734  | -0.00741 | -0.00767 |
|                                  | (A0 * !B * Y)       | 0.01328   | 0.01351  | 0.01323  |
|                                  | (A0 * !B * Y)       | -0.00651  | -0.00651 | -0.00649 |
|                                  | (!A0 * A2 * B * !Y) | 0.00472   | 0.00475  | 0.00435  |
|                                  | (!A0 * A2 * B * !Y) | -0.00667  | -0.00674 | -0.00739 |
|                                  | (!A0 * !B * Y)      | 0.01253   | 0.01267  | 0.01256  |
|                                  | (!A0 * !B * Y)      | -0.00650  | -0.00650 | -0.00649 |

Passive power(pJ) for A2 rising (conditional):

| Cell Name                        | When                                        | Power(pJ) |          |          |
|----------------------------------|---------------------------------------------|-----------|----------|----------|
|                                  |                                             | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__oai31_1 | $(A0 * A1 * B * !Y)$                        | -0.01313  | -0.01340 | -0.01329 |
|                                  | $(A0 * A1 * B * !Y)$                        | 0.00655   | 0.00655  | 0.00652  |
|                                  | $(A0 * !B * Y)$                             | -0.01316  | -0.01341 | -0.01326 |
|                                  | $(A0 * !B * Y)$                             | 0.00652   | 0.00655  | 0.00652  |
|                                  | $(A0 * !A1 * B * !Y) + (!A0 * A1 * B * !Y)$ | -0.01296  | -0.01341 | -0.01327 |
|                                  | $(A0 * !A1 * B * !Y) + (!A0 * A1 * B * !Y)$ | 0.00675   | 0.00655  | 0.00652  |
|                                  | $(!A0 * A1 * !B * Y)$                       | -0.01221  | -0.01313 | -0.01288 |
|                                  | $(!A0 * A1 * !B * Y)$                       | 0.00654   | 0.00670  | 0.00652  |
|                                  | $(!A0 * !A1 * !B * Y)$                      | -0.01351  | -0.01353 | -0.01351 |
|                                  | $(!A0 * !A1 * !B * Y)$                      | 0.00644   | 0.00650  | 0.00644  |

Passive power(pJ) for A2 falling (conditional):

| Cell Name                        | When                                        | Power(pJ) |          |          |
|----------------------------------|---------------------------------------------|-----------|----------|----------|
|                                  |                                             | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__oai31_1 | $(A0 * A1 * B * !Y)$                        | 0.01340   | 0.01340  | 0.01329  |
|                                  | $(A0 * A1 * B * !Y)$                        | -0.00650  | -0.00650 | -0.00649 |
|                                  | $(A0 * !B * Y)$                             | 0.01329   | 0.01341  | 0.01326  |
|                                  | $(A0 * !B * Y)$                             | -0.00650  | -0.00649 | -0.00649 |
|                                  | $(A0 * !A1 * B * !Y) + (!A0 * A1 * B * !Y)$ | 0.01329   | 0.01341  | 0.01327  |
|                                  | $(A0 * !A1 * B * !Y) + (!A0 * A1 * B * !Y)$ | -0.00649  | -0.00649 | -0.00649 |
|                                  | $(!A0 * A1 * !B * Y)$                       | 0.01290   | 0.01313  | 0.01288  |
|                                  | $(!A0 * A1 * !B * Y)$                       | -0.00648  | -0.00650 | -0.00649 |
|                                  | $(!A0 * !A1 * !B * Y)$                      | 0.01367   | 0.01359  | 0.01357  |
|                                  | $(!A0 * !A1 * !B * Y)$                      | -0.00629  | -0.00647 | -0.00644 |

Passive power(pJ) for B rising (conditional):

| Cell Name                        | When                  | Power(pJ) |          |          |
|----------------------------------|-----------------------|-----------|----------|----------|
|                                  |                       | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__oai31_1 | (!A0 * !A1 * !A2 * Y) | -0.01386  | -0.01384 | -0.01405 |
|                                  | (!A0 * !A1 * !A2 * Y) | 0.00193   | 0.00193  | 0.00177  |

Passive power(pJ) for B falling (conditional):

| Cell Name                        | When                  | Power(pJ) |          |          |
|----------------------------------|-----------------------|-----------|----------|----------|
|                                  |                       | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__oai31_1 | (!A0 * !A1 * !A2 * Y) | 0.01420   | 0.01426  | 0.01420  |
|                                  | (!A0 * !A1 * !A2 * Y) | -0.00168  | -0.00168 | -0.00167 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_OR2\_1

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT |   | OUTPUT |
|-------|---|--------|
| A     | B | Y      |
| 0     | 0 | 0      |
| x     | 1 | 1      |
| 1     | x | 1      |

## Footprint

| Cell Name                      | Area     |
|--------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__or2_1 | 23.94000 |

## Pin Capacitance Information

| Cell Name                      | Pin Cap(pf) |         | Max Cap(pf) |
|--------------------------------|-------------|---------|-------------|
|                                | A           | B       | Y           |
| gf180mcu_osu_sc_gp9t3v3__or2_1 | 0.00405     | 0.00399 | 15.52995    |

## Leakage Information

| Cell Name                      | Leakage(nW) |         |         |
|--------------------------------|-------------|---------|---------|
|                                | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__or2_1 | 0.00000     | 0.00164 | 0.00237 |

## Delay Information

Delay(ns) to Y rising :

| Cell Name                      | Timing Arc(Dir) | Delay(ns) |         |          |
|--------------------------------|-----------------|-----------|---------|----------|
|                                |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__or2_1 | A->Y (RR)       | 0.07704   | 1.35185 | 59.55870 |
|                                | B->Y (RR)       | 0.09158   | 1.68722 | 65.55750 |

Delay(ns) to Y falling :

| Cell Name                      | Timing Arc(Dir) | Delay(ns) |         |          |
|--------------------------------|-----------------|-----------|---------|----------|
|                                |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__or2_1 | A->Y (FF)       | 0.11254   | 2.77114 | 81.20590 |
|                                | B->Y (FF)       | 0.12740   | 2.51327 | 76.95490 |



## Power Information

Internal switching power(pJ) to Y rising :

| Cell Name                      | Input | Power(pJ) |         |         |
|--------------------------------|-------|-----------|---------|---------|
|                                |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__or2_1 | A     | 0.01533   | 0.30532 | 5.50672 |
|                                | A     | 0.03779   | 0.32771 | 5.50706 |
|                                | B     | 0.02127   | 0.36661 | 6.53283 |
|                                | B     | 0.05323   | 0.39815 | 6.55744 |

Internal switching power(pJ) to Y falling :

| Cell Name                      | Input | Power(pJ) |         |         |
|--------------------------------|-------|-----------|---------|---------|
|                                |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__or2_1 | A     | 0.03943   | 0.33171 | 5.49119 |
|                                | A     | 0.01684   | 0.30994 | 5.47204 |
|                                | B     | 0.04831   | 0.38678 | 6.53810 |
|                                | B     | 0.01629   | 0.35535 | 6.51616 |

Passive power(pJ) for A rising (conditional):

| Cell Name                      | When    | Power(pJ) |          |          |
|--------------------------------|---------|-----------|----------|----------|
|                                |         | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__or2_1 | (B * Y) | -0.00437  | -0.00445 | -0.00434 |
|                                | (B * Y) | 0.00781   | 0.00800  | 0.00781  |

Passive power(pJ) for A falling (conditional):

| Cell Name                      | When    | Power(pJ) |          |          |
|--------------------------------|---------|-----------|----------|----------|
|                                |         | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__or2_1 | (B * Y) | 0.00470   | 0.00468  | 0.00448  |
|                                | (B * Y) | -0.00742  | -0.00741 | -0.00767 |

Passive power(pJ) for B rising (conditional):

| Cell Name                      | When    | Power(pJ) |          |          |
|--------------------------------|---------|-----------|----------|----------|
|                                |         | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__or2_1 | (A * Y) | -0.01267  | -0.01342 | -0.01328 |
|                                | (A * Y) | 0.00716   | 0.00655  | 0.00652  |

Passive power(pJ) for B falling (conditional):

| Cell Name                      | When    | Power(pJ) |          |          |
|--------------------------------|---------|-----------|----------|----------|
|                                |         | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__or2_1 | (A * Y) | 0.01340   | 0.01342  | 0.01328  |
|                                | (A * Y) | -0.00649  | -0.00647 | -0.00647 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_TBUF\_1

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT |    | OUTPUT |
|-------|----|--------|
| A     | EN | Y      |
| -     | 0  | HiZ    |
| 0     | 1  | 0      |
| 1     | 1  | 1      |

## Footprint

| Cell Name                       | Area     |
|---------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__tbuf_1 | 34.02000 |

## Pin Capacitance Information

| Cell Name                       | Pin Cap(pf) |         | Max Cap(pf) |
|---------------------------------|-------------|---------|-------------|
|                                 | A           | EN      | Y           |
| gf180mcu_osu_sc_gp9t3v3__tbuf_1 | 0.00404     | 0.00535 | 8.22652     |

## Leakage Information

| Cell Name                       | Leakage(nW) |         |         |
|---------------------------------|-------------|---------|---------|
|                                 | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__tbuf_1 | 0.00000     | 0.00183 | 0.00201 |

## Delay Information

Delay(ns) to Y rising :

| Cell Name                       | Timing Arc(Dir) | Delay(ns) |         |          |
|---------------------------------|-----------------|-----------|---------|----------|
|                                 |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__tbuf_1 | A->Y (RR)       | 0.12619   | 2.11138 | 64.48500 |
|                                 | EN->Y (FR)      | 0.06098   | 3.59158 | 65.65660 |
|                                 | EN->Y (RR)      | 0.07380   | 2.12059 | 66.82990 |

Delay(ns) to Y falling :

| Cell Name                       | Timing Arc(Dir) | Delay(ns) |         |          |
|---------------------------------|-----------------|-----------|---------|----------|
|                                 |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__tbuf_1 | A->Y (FF)       | 0.11778   | 2.31779 | 60.81760 |
|                                 | EN->Y (FF)      | 0.07384   | 3.59158 | 65.65660 |
|                                 | EN->Y (RF)      | 0.02527   | 2.28278 | 70.40980 |

## Power Information

Internal switching power(pJ) to Y rising :

| Cell Name                       | Input | Power(pJ) |         |         |
|---------------------------------|-------|-----------|---------|---------|
|                                 |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__tbuf_1 | A     | 0.03278   | 0.40406 | 6.96116 |
|                                 | A     | 0.04954   | 0.42046 | 6.97209 |
|                                 | EN    | 0.01584   | 0.38154 | 6.91209 |
|                                 | EN    | 0.03897   | 0.40460 | 6.93339 |

Internal switching power(pJ) to Y falling :

| Cell Name                       | Input | Power(pJ) |         |         |
|---------------------------------|-------|-----------|---------|---------|
|                                 |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__tbuf_1 | A     | 0.04280   | 0.41656 | 6.93949 |
|                                 | A     | 0.02605   | 0.40010 | 6.93222 |
|                                 | EN    | 0.00851   | 0.37860 | 6.91171 |
|                                 | EN    | 0.03758   | 0.40761 | 6.94411 |

Passive power(pJ) for A rising (conditional):

| Cell Name                       | When | Power(pJ) |         |         |
|---------------------------------|------|-----------|---------|---------|
|                                 |      | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__tbuf_1 | !EN  | 0.00892   | 0.37958 | 6.86939 |
|                                 | !EN  | 0.03087   | 0.40163 | 6.89135 |

Passive power(pJ) for A falling (conditional):

| Cell Name                       | When | Power(pJ) |         |         |
|---------------------------------|------|-----------|---------|---------|
|                                 |      | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__tbuf_1 | !EN  | 0.02631   | 0.39882 | 6.88807 |
|                                 | !EN  | 0.00426   | 0.37674 | 6.86607 |

Passive power(pJ) for EN rising (conditional):

| Cell Name                       | When      | Power(pJ) |         |         |
|---------------------------------|-----------|-----------|---------|---------|
|                                 |           | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__tbuf_1 | (A * Y)   | 0.00798   | 0.38131 | 6.87217 |
|                                 | (A * Y)   | 0.03227   | 0.40585 | 6.89660 |
|                                 | (!A * !Y) | 0.00014   | 0.37296 | 6.86687 |
|                                 | (!A * !Y) | 0.02855   | 0.40152 | 6.89527 |

Passive power(pJ) for EN falling (conditional):

| Cell Name                       | When      | Power(pJ) |         |         |
|---------------------------------|-----------|-----------|---------|---------|
|                                 |           | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__tbuf_1 | (A * Y)   | 0.02073   | 0.39493 | 6.88428 |
|                                 | (A * Y)   | -0.00367  | 0.37039 | 6.85986 |
|                                 | (!A * !Y) | 0.02109   | 0.39932 | 6.88887 |
|                                 | (!A * !Y) | -0.00728  | 0.37081 | 6.86034 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_TIEH

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Footprint

| Cell Name                     | Area     |
|-------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__tieh | 15.12000 |

## Pin Capacitance Information

| Cell Name                     | Max Cap(pf) |
|-------------------------------|-------------|
|                               | Y           |
| gf180mcu_osu_sc_gp9t3v3__tieh | 34.42027    |

## Leakage Information

| Cell Name                     | Leakage(nW) |         |         |
|-------------------------------|-------------|---------|---------|
|                               | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__tieh | 0.00000     | 0.00000 | 0.00000 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_TIEL

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

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## Footprint

| Cell Name                     | Area     |
|-------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__tiel | 15.12000 |

## Pin Capacitance Information

| Cell Name                     | Max Cap(pf) |
|-------------------------------|-------------|
|                               | Y           |
| gf180mcu_osu_sc_gp9t3v3__tiel | 51.62853    |

## Leakage Information

| Cell Name                     | Leakage(nW) |         |         |
|-------------------------------|-------------|---------|---------|
|                               | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__tiel | 0.00000     | 0.00000 | 0.00000 |



# GF180MCU\_OSU\_SC\_GP9T3V3\_\_TINV\_1

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT |    | OUTPUT |
|-------|----|--------|
| A     | EN | Y      |
| -     | 0  | HiZ    |
| 0     | 1  | 1      |
| 1     | 1  | 0      |

## Footprint

| Cell Name                       | Area     |
|---------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__tinv_1 | 24.57000 |

## Pin Capacitance Information

| Cell Name                       | Pin Cap(pf) |         | Max Cap(pf) |
|---------------------------------|-------------|---------|-------------|
|                                 | A           | EN      | Y           |
| gf180mcu_osu_sc_gp9t3v3__tinv_1 | 0.00396     | 0.00535 | 8.01146     |

## Leakage Information

| Cell Name                       | Leakage(nW) |         |         |
|---------------------------------|-------------|---------|---------|
|                                 | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__tinv_1 | 0.00000     | 0.00109 | 0.00144 |

## Delay Information

Delay(ns) to Y rising :

| Cell Name                       | Timing Arc(Dir) | Delay(ns) |         |          |
|---------------------------------|-----------------|-----------|---------|----------|
|                                 |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__tinv_1 | A->Y (FR)       | 0.08888   | 3.26317 | 86.79460 |
|                                 | EN->Y (FR)      | 0.06098   | 3.59158 | 65.65660 |
|                                 | EN->Y (RR)      | 0.07386   | 2.08766 | 65.01830 |

Delay(ns) to Y falling :

| Cell Name                       | Timing Arc(Dir) | Delay(ns) |         |          |
|---------------------------------|-----------------|-----------|---------|----------|
|                                 |                 | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__tinv_1 | A->Y (RF)       | 0.06768   | 2.19253 | 61.92080 |
|                                 | EN->Y (FF)      | 0.07396   | 3.59158 | 65.65660 |
|                                 | EN->Y (RF)      | 0.02528   | 2.24631 | 68.90440 |

## Power Information

Internal switching power(pJ) to Y rising :

| Cell Name                       | Input | Power(pJ) |         |         |
|---------------------------------|-------|-----------|---------|---------|
|                                 |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__tinv_1 | A     | 0.03704   | 0.20704 | 2.49375 |
|                                 | A     | 0.01040   | 0.18006 | 2.46488 |
|                                 | EN    | 0.01578   | 0.38175 | 6.91353 |
|                                 | EN    | 0.03849   | 0.40431 | 6.93032 |

Internal switching power(pJ) to Y falling :

| Cell Name                       | Input | Power(pJ) |         |         |
|---------------------------------|-------|-----------|---------|---------|
|                                 |       | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__tinv_1 | A     | 0.00162   | 0.16032 | 2.24365 |
|                                 | A     | 0.02818   | 0.18736 | 2.27567 |
|                                 | EN    | 0.00758   | 0.37743 | 6.90886 |
|                                 | EN    | 0.03756   | 0.40776 | 6.94765 |

Passive power(pJ) for A rising (conditional):

| Cell Name                       | When | Power(pJ) |          |          |
|---------------------------------|------|-----------|----------|----------|
|                                 |      | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__tinv_1 | !EN  | -0.01329  | -0.01352 | -0.01341 |
|                                 | !EN  | 0.00654   | 0.00649  | 0.00641  |

Passive power(pJ) for A falling (conditional):

| Cell Name                       | When | Power(pJ) |          |          |
|---------------------------------|------|-----------|----------|----------|
|                                 |      | first     | mid      | last     |
| gf180mcu_osu_sc_gp9t3v3__tinv_1 | !EN  | 0.01353   | 0.01363  | 0.01341  |
|                                 | !EN  | -0.00628  | -0.00649 | -0.00641 |

Passive power(pJ) for EN rising (conditional):

| Cell Name                       | When     | Power(pJ) |         |         |
|---------------------------------|----------|-----------|---------|---------|
|                                 |          | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__tinv_1 | (A * !Y) | -0.00010  | 0.37270 | 6.86663 |
|                                 | (A * !Y) | 0.02854   | 0.40151 | 6.89527 |
|                                 | (!A * Y) | 0.00798   | 0.38130 | 6.87217 |
|                                 | (!A * Y) | 0.03216   | 0.40573 | 6.89651 |

Passive power(pJ) for EN falling (conditional):

| Cell Name                       | When     | Power(pJ) |         |         |
|---------------------------------|----------|-----------|---------|---------|
|                                 |          | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__tinv_1 | (A * !Y) | 0.02128   | 0.39951 | 6.88906 |
|                                 | (A * !Y) | -0.00729  | 0.37081 | 6.86033 |
|                                 | (!A * Y) | 0.02067   | 0.39491 | 6.88428 |
|                                 | (!A * Y) | -0.00359  | 0.37045 | 6.85994 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_XNOR2\_1

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT |   | OUTPUT |
|-------|---|--------|
| A     | B | Y      |
| 0     | 0 | 1      |
| 0     | 1 | 0      |
| 1     | 0 | 0      |
| 1     | 1 | 1      |

## Footprint

| Cell Name                        | Area     |
|----------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__xnor2_1 | 41.58000 |

## Pin Capacitance Information

| Cell Name                        | Pin Cap(pf) |         | Max Cap(pf) |
|----------------------------------|-------------|---------|-------------|
|                                  | A           | B       | Y           |
| gf180mcu_osu_sc_gp9t3v3__xnor2_1 | 0.00806     | 0.00800 | 7.98787     |

## Leakage Information

| Cell Name                        | Leakage(nW) |         |         |
|----------------------------------|-------------|---------|---------|
|                                  | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__xnor2_1 | 0.00000     | 0.00284 | 0.00353 |

## Delay Information

Delay(ns) to Y rising (conditional):

| Cell Name                        | Timing Arc(Dir) | When | Delay(ns) |         |          |
|----------------------------------|-----------------|------|-----------|---------|----------|
|                                  |                 |      | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__xnor2_1 | A->Y (RR)       | B    | 0.11993   | 2.06957 | 62.47440 |
|                                  | A->Y (FR)       | !B   | 0.08486   | 3.89962 | 98.43540 |
|                                  | B->Y (RR)       | A    | 0.09887   | 2.14440 | 64.92940 |
|                                  | B->Y (FR)       | !A   | 0.09569   | 3.27339 | 86.63570 |

Delay(ns) to Y falling (conditional):

| Cell Name                        | Timing Arc(Dir) | When | Delay(ns) |         |          |
|----------------------------------|-----------------|------|-----------|---------|----------|
|                                  |                 |      | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__xnor2_1 | A->Y (FF)       | B    | 0.13096   | 2.41292 | 62.00250 |
|                                  | A->Y (RF)       | !B   | 0.05707   | 2.11559 | 61.20550 |
|                                  | B->Y (FF)       | A    | 0.10249   | 2.38066 | 61.95390 |
|                                  | B->Y (RF)       | !A   | 0.07502   | 2.20292 | 61.80200 |

## Power Information

Internal switching power(pJ) to Y rising (conditional):

| Cell Name                        | Input | When | Power(pJ) |         |         |
|----------------------------------|-------|------|-----------|---------|---------|
|                                  |       |      | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__xnor2_1 | A     | B    | 0.02341   | 0.38903 | 6.93915 |
|                                  | A     | B    | 0.05618   | 0.42141 | 6.96092 |
|                                  | A     | !B   | 0.04955   | 0.56372 | 8.99276 |
|                                  | A     | !B   | 0.00540   | 0.51943 | 8.94816 |
|                                  | B     | A    | 0.00513   | 0.37168 | 6.90723 |
|                                  | B     | A    | 0.04557   | 0.41194 | 6.94134 |
|                                  | B     | !A   | 0.05892   | 0.59841 | 9.43520 |
|                                  | B     | !A   | 0.00524   | 0.54423 | 9.37954 |

Internal switching power(pJ) to Y falling (conditional):

| Cell Name                        | Input | When | Power(pJ) |         |         |
|----------------------------------|-------|------|-----------|---------|---------|
|                                  |       |      | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__xnor2_1 | A     | B    | 0.05931   | 0.42532 | 6.95424 |
|                                  | A     | B    | 0.02817   | 0.39430 | 6.92305 |
|                                  | A     | !B   | 0.01411   | 0.51897 | 8.92059 |
|                                  | A     | !B   | 0.05786   | 0.56309 | 8.96430 |
|                                  | B     | A    | 0.05442   | 0.42241 | 6.94532 |
|                                  | B     | A    | 0.01389   | 0.38188 | 6.90813 |
|                                  | B     | !A   | 0.01515   | 0.54069 | 9.15520 |
|                                  | B     | !A   | 0.06829   | 0.59431 | 9.21313 |

# GF180MCU\_OSU\_SC\_GP9T3V3\_\_XOR2\_1

gf180mcu\_osu\_sc\_gp9t3v3\_TT\_25C.ccs  
Cell Library: Process , Voltage 3.30,  
Temp 25.00

## Truth Table

| INPUT |   | OUTPUT |
|-------|---|--------|
| A     | B | Y      |
| 0     | 0 | 0      |
| 0     | 1 | 1      |
| 1     | 0 | 1      |
| 1     | 1 | 0      |

## Footprint

| Cell Name                       | Area     |
|---------------------------------|----------|
| gf180mcu_osu_sc_gp9t3v3__xor2_1 | 43.47000 |

## Pin Capacitance Information

| Cell Name                       | Pin Cap(pf) |         | Max Cap(pf) |
|---------------------------------|-------------|---------|-------------|
|                                 | A           | B       | Y           |
| gf180mcu_osu_sc_gp9t3v3__xor2_1 | 0.00800     | 0.00802 | 7.96151     |

## Leakage Information

| Cell Name                       | Leakage(nW) |         |         |
|---------------------------------|-------------|---------|---------|
|                                 | Min.        | Avg     | Max.    |
| gf180mcu_osu_sc_gp9t3v3__xor2_1 | 0.00000     | 0.00284 | 0.00327 |



## Delay Information

Delay(ns) to Y rising (conditional):

| Cell Name                       | Timing Arc(Dir) | When | Delay(ns) |         |          |
|---------------------------------|-----------------|------|-----------|---------|----------|
|                                 |                 |      | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__xor2_1 | A->Y (RR)       | !B   | 0.09898   | 2.14029 | 64.70250 |
|                                 | A->Y (FR)       | B    | 0.09767   | 3.26650 | 86.42150 |
|                                 | B->Y (RR)       | !A   | 0.12649   | 2.16480 | 64.72720 |
|                                 | B->Y (FR)       | A    | 0.07978   | 3.19471 | 85.83090 |

Delay(ns) to Y falling (conditional):

| Cell Name                       | Timing Arc(Dir) | When | Delay(ns) |         |          |
|---------------------------------|-----------------|------|-----------|---------|----------|
|                                 |                 |      | First     | Mid     | Last     |
| gf180mcu_osu_sc_gp9t3v3__xor2_1 | A->Y (FF)       | !B   | 0.10243   | 2.37681 | 61.80080 |
|                                 | A->Y (RF)       | B    | 0.07340   | 2.19913 | 61.63550 |
|                                 | B->Y (FF)       | !A   | 0.10701   | 2.27481 | 59.14490 |
|                                 | B->Y (RF)       | A    | 0.07289   | 2.77028 | 73.56620 |

## Power Information

Internal switching power(pJ) to Y rising (conditional):

| Cell Name                       | Input | When | Power(pJ) |         |         |
|---------------------------------|-------|------|-----------|---------|---------|
|                                 |       |      | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__xor2_1 | A     | B    | 0.06424   | 0.60267 | 9.44541 |
|                                 | A     | B    | 0.01553   | 0.55363 | 9.39487 |
|                                 | A     | !B   | 0.00372   | 0.37039 | 6.91341 |
|                                 | A     | !B   | 0.04495   | 0.41132 | 6.94768 |
|                                 | B     | A    | 0.05745   | 0.58074 | 9.24599 |
|                                 | B     | A    | 0.01376   | 0.53639 | 9.20000 |
|                                 | B     | !A   | 0.01360   | 0.37756 | 6.91995 |
|                                 | B     | !A   | 0.04963   | 0.41353 | 6.94914 |

Internal switching power(pJ) to Y falling (conditional):

| Cell Name                       | Input | When | Power(pJ) |         |         |
|---------------------------------|-------|------|-----------|---------|---------|
|                                 |       |      | first     | mid     | last    |
| gf180mcu_osu_sc_gp9t3v3__xor2_1 | A     | B    | 0.00911   | 0.53470 | 9.15333 |
|                                 | A     | B    | 0.05846   | 0.58467 | 9.20271 |
|                                 | A     | !B   | 0.05571   | 0.42355 | 6.95424 |
|                                 | A     | !B   | 0.01457   | 0.38260 | 6.91307 |
|                                 | B     | A    | 0.00960   | 0.52688 | 8.87904 |
|                                 | B     | A    | 0.05392   | 0.57170 | 8.92334 |
|                                 | B     | !A   | 0.06051   | 0.42874 | 6.95863 |
|                                 | B     | !A   | 0.02344   | 0.39192 | 6.92336 |